

FOUNDATIONS AND COMPUTATION

Computable Surreal Numbers

Structural names, Puiseux series,
and effective left-finite fields

A unified mathematical report with proofs,
representation contracts, and a reconciliation ledger

Merged and reconciled from three supplied reports

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Central distinction. A program describing a well-founded surreal construction need not compute its real value in the ordinary numerical sense. Conversely, a coefficient algorithm need not reveal the finite support information needed for multiplication. This article develops both notions without identifying them.

Status. Common material is consolidated; distinct representations and complementary proofs are retained. The reconciliation ledger records source locations, strengthened formulations, and corrections. Established results remain attributed to prior work; no new priority claim is made. The article has not been refereed or formalized in Lean. Its accompanying finite exact-arithmetic checks are tests, not machine-checked proofs.

Abstract

This report merges three accounts of computable surreal numbers into one representation-sensitive theory. Hereditarily effective Conway-cut names define the established structural field $\mathcal{S}_{\text{str}}$, whose real trace is the hyperarithmetic reals. For ordinary numerical access, two different fields must be retained: $\mathcal{P}_c$, using a lower-bounded lattice with one ramification denominator per name, and $\mathcal{L}_c$, using rational left-finite supports equipped with computable finite candidate covers. Both are countable real-closed subfields of $\mathbf{No}$ with real trace exactly $\mathbb{R}_c$, and $\mathcal{P}_c \subsetneq \mathcal{L}_c$. The larger field is the effective sequential completion of the smaller for a supplied exact valuation stabilization modulus; neither assertion is an unqualified classical completeness claim.

The unified proofs give uniform ring operations, exact symbolic truncation, certified reciprocal and root algorithms, effective Hensel lifting, formal functional calculus, and finite-rank extensions. Two independent routes to Puiseux real closedness are preserved: a henselian valuation argument and a Newton–Puiseux argument with an explicit finite-jet lifting lemma. The latter explains why finitely many branch data suffice for a single coefficient algorithm, without assuming a uniform branch selector.

The obstructions are equally explicit. Computable Hahn coefficients can have a noncomputable product, even with decidable left-finite supports; a separate example has every convolution fiber of size at most one. One halting oracle is sharp for unrestricted coefficient convolution and for several normalization tasks, but its output role differs in the two settings. Equality, sign, inversion, roots, and modulus are not silently conflated with extensional closure. Oracle-relative fields recover the Turing-degree ordering. A source crosswalk, a corrected nonzero-polynomial hypothesis, repository-specific implementation contracts, and independently rerun finite checks complete the reconciliation. None of the finite tests is a formal proof of the infinite mathematical statements.

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1 The question and the reconciled answer

A computable real number is not a real number whose entire decimal expansion has already been stored. It is one for which a finite algorithm can supply arbitrarily accurate rational approximations with an effective error bound. Replacing $\mathbb{R}$ by **No** changes much more than the possible magnitude of the answer. A surreal has an ordinal-length sign sequence and a transfinite normal form. There is no single sequence of positive surreal error tolerances cofinal toward zero in the whole class. Moreover, the algebra of infinite normal forms depends on support information that individual coefficient queries need not provide.

The appropriate conclusion is not that computation is impossible, but that *the representation is part of the mathematical definition*. We first separate structural from numerical effectivity:

$$\mathcal{S}_{\text{str}} = \text{structurally computable surreals}, \quad \mathcal{L}_c = \text{numerically computable left-finite surreals}.$$

The first is broad and hereditary. The second is a precisely specified rank-one numerical workspace containing the distinct Puiseux model $\mathcal{P}_c$. None is advertised as the unique possible meaning of “computable surreal.”

1.1 Source identities and the central reconciliation

Throughout, **A** denotes the report in `computable_surreal_numbers.zip`; **B** the report in `computable_surreal_numbers(1).zip`; and **C** the report in `computable_surreal_numbers(2).zip`. The suffixes are upload identifiers, not chronological or mathematical version numbers. Their source manuscripts are *not* redistributed with this report; the accompanying provenance manifest records their checksums and line counts. Appendix B provides the topic and correction crosswalk.

A uses a larger effective left-finite field as its main numerical model; B and C use a bounded-denominator Puiseux field. They therefore do not make inconsistent assertions about one field. We keep both symbols and prove the inclusion explicitly. In particular, C’s request for an effective support enlargement beyond Puiseux series is answered, at the left-finite level, by A’s finite-cover construction. The question left open here concerns more general supports, not that already constructed enlargement.

The strongest compatible statements have been selected rather than averaged: C’s full validity classification and degree-preserving sign example; A’s left-finite support counterexample, Hensel construction, and effective completion; B’s sharp convolution oracle bound, algebraic-branch obstruction, and differential consequences; and B–C’s finite-jet argument and finite-rank Puiseux construction. C’s oracle hierarchy is extended below to both numerical models by the same explicit proof. These are synthesis choices, not assertions of independent novelty.

Model	Information supplied by a name	Real trace	Main qualification
$\mathcal{S}_{\text{str}}$	Hereditary enumerations of separated cuts	Hyperarithmetical reals	Semantic validity is Π_1^1 -complete.
$\mathcal{P}_c$	Uniform real coefficients on one lower-bounded rational lattice	$\mathbb{R}_c$	Computable-modulus limits may escape the lattice.
$\mathcal{L}_c$	Uniform real coefficients and complete finite support covers	$\mathbb{R}_c$	Effective completeness requires a supplied modulus.

Both numerical models have uniform ring operations but not a uniform raw-name inverse. Neither real trace nor algebraic closedness supplies a missing representation converter. We do not assert a general computable translation between structural and numerical names.

1.2 Prior work and what is being developed

Recursive surreal numbers were studied by Harkleroad in 1990 and several notions were compared by Lurie in 1998 [3, 4]. Later presentations by Hamkins, reporting joint work with Turetsky, explicitly identify the hereditary structural version with hyperarithmetical sign sequences and establish its real closedness [5, 13, 6]. Those results are *prior theory*, not discoveries of this article.

The numerical carrier used below is equally classical: real series with rational, left-finite support form the Levi–Civita field. A recent primary source recalls this definition and its classical algebraic properties [8, §1.1]. Our question is which of its elements and operations are effective under a representation that continues to accept ordinary Cauchy names for real constants. The effective finite-cover requirement, the explicit counterexamples, and the resulting collection of closure and nonuniformity proofs are the focus here. We do not infer novelty merely from having supplied a proof; related effective valued-field formulations may exist under other representations or terminology.

The standard background is Conway’s construction and Gonshor’s normal-form theory [1, 2]. Only one substantial valuation-theoretic input is used in the numerical real-closedness proof: a henselian finite extension in residue characteristic zero has degree equal to ramification index times residue degree. We isolate that input explicitly in Section 8.

The independent finite-jet proof in Section 10 instead uses the characteristic-zero Newton–Puiseux theorem, stated explicitly in the introductions of [14, 15]. Its additional effective step is proved, not inferred from the computability of each coefficient separately. The user-supplied overview [18] motivates the ordinary-real baseline; the report fixes fast Cauchy names rather than treating decimal, Dedekind-cut, and Cauchy encodings as uniformly interchangeable. Objectwise equivalence of computable values is weaker than a uniform conversion of names.

All algorithms here are ordinary finite-time Turing computations per query. The ordinal length of a support is not a machine runtime. Infinitary-machine approaches, such as the κ -time and κ -tape framework of Galeotti–Nobrega [16], change the computational model and are not counterexamples to the ordinary-computability obstructions below.

1.3 A map of the main results

The main numerical theorem, assembled from the proofs below, is as follows.

Theorem 1.1 (Numerically conservative effective core). *Let $\mathcal{L}_c$ be the class of surreal normal forms with an effective left-finite Cauchy–Hahn name, as in Definition 6.1. Then:*

- (i) $\mathcal{L}_c$ is a countable real-closed subfield of $\mathbf{No}$, and $\mathcal{L}_c \cap \mathbb{R} = \mathbb{R}_c$. Its natural valuation has value group $\mathbb{Q}$ and residue field $\mathbb{R}_c$.
- (ii) Addition, subtraction, multiplication, rational monomial shifts, and coefficient extraction are computable uniformly on names.
- (iii) Every nonzero element has an inverse in $\mathcal{L}_c$. Computing an inverse uniformly is possible when the leading exponent is supplied, and is impossible on all nonzero raw names without additional information.
- (iv) Simple residue roots lift effectively. The computable Puiseux field $\mathcal{P}_c$ is a proper real-closed subfield of $\mathcal{L}_c$, and $\mathcal{L}_c$ is its effective sequential completion for exact valuation stabilization.
- (v) $\mathcal{L}_c[i]$ is algebraically closed. Formal evaluation at infinitesimals, formal differentiation, and residue-free formal integration admit explicit algorithms.

This theorem does *not* say that $\mathcal{L}_c$, with its raw representation, is a computable ordered field in the sense of having decidable equality and uniform field operations. Indeed, part (iii) rules out that interpretation. The distinction between an extensional algebraic closure property and a uniform operation on names will be maintained throughout.

Two negative results explain why the qualifications in this theorem are substantive. Theorem 5.1 constructs computable coefficient functions, even with decidable supports, whose Hahn product has a noncomputable coefficient function. Theorem 12.2 shows that uniform inversion already conflicts with ordinary real input and coefficient access on the rational-function family $a + t$.

2 Conventions, names, and three topologies

2.1 Surreal and Hahn conventions

We work classically, for example in ZFC with definable classes, or in a universe arrangement in which the full surreal carrier is one universe larger than the small option families. Every individual support and every program considered here is a set. We never quantify over a purported set of all surreals.

For small separated sets $L, R \subseteq \mathbf{No}$, $\{L \mid R\}$ denotes the *simplest* surreal strictly between every member of L and every member of R . It is not an ordinary supremum. The birthday $b(x)$ is the length of the canonical sign sequence of x .

Fix

$$t = \omega^{-1}, \quad t^q = \omega^{-q} \quad (q \in \mathbb{Q}).$$

Under Conway normal forms, the usual Hahn field $\mathbb{R}((t^{\mathbb{Q}}))$, with well-ordered support in the increasing exponent convention, is an ordered subfield of $\mathbf{No}$. All uses of this identification are

classical consequences of normal-form theory [1, 2]; we do not claim that the repository already formalizes this bridge. A series is written

$$x = \sum_{q \in \mathbb{Q}} a_q t^q, \quad v(x) = \min \text{supp}(x) \quad (x \neq 0), \quad v(0) = +\infty.$$

Its sign is the sign of $a_{v(x)}$. Larger valuation means a smaller non-Archimedean scale. We set $[t^q]x = a_q$.

A set $S \subseteq \mathbb{Q}$ is *left-finite* if $S \cap (-\infty, B)$ is finite for every rational B . Every nonempty left-finite set is bounded below and has a least element: choose one member s , and inspect the finite nonempty set $S \cap (-\infty, s + 1)$. The same argument applies to each nonempty subset of S , so S is well ordered. An infinite left-finite support has order type ω and tends to $+\infty$.

2.2 What a computable real coefficient means

Write $\mathbb{R}_c$ for the ordinary computable reals. Our real names are functions $A : \mathbb{N} \rightarrow \mathbb{Q}$ satisfying

$$|A(k) - a| \leq 2^{-k}. \quad (2.1)$$

For a family a_q we require a *single* algorithm $A(q, k)$ satisfying this bound jointly in the rational index q and precision k . Saying merely that each a_q is a computable real is weaker: the family $a_n = \chi_K(n)$ consists of rational numbers but is not uniformly computable. Here K is a fixed undecidable computably enumerable halting set.

A name is a finite program plus any explicitly specified finite data. Correctness conditions on a name are semantic promises. There is no assumption that an interpreter can verify totality, all error inequalities, or all support conditions. A *uniform operation* is an algorithm which transforms every valid input name satisfying its stated promises into a valid output name. An *extensional closure theorem* only asserts that every resulting value has some computable name.

2.3 Why full-surreal Cauchy approximation is the wrong starting point

Proposition 2.1 (No small nontrivial convergence in the fine order topology). *A set-indexed net in $\mathbf{No}$ which converges in the full order topology is eventually equal to its limit. A set-indexed Cauchy net for its ordered-field uniformity is eventually constant.*

Proof. For a net $(x_i)_{i \in I}$ and a proposed limit x , form the set $D = \{|x_i - x| : i \in I, x_i \neq x\}$ of positive distances. The small cut $\varepsilon = \{0 \mid D\}$ is positive and smaller than every member of D . Convergence eventually gives $|x_i - x| < \varepsilon$, forcing $x_i = x$. For the Cauchy assertion use instead the set of all nonzero pairwise distances. The Cauchy condition makes all sufficiently late pairwise differences zero. Empty distance sets cause no difficulty. $\square$

This is consistent with the topology results documented in the repository [12]. It does not prevent computation in a fixed workspace. Inside the Levi–Civita field, t^n tends to zero for its *intrinsic* valuation topology, since n eventually exceeds every rational valuation. In $\mathbf{No}$, however, $t^n > \omega^{-\omega}$ for every integer n . Indeed, the subspace topology that the full surreal order induces on the rational-exponent field is discrete, whereas its intrinsic order topology is not.

There are thus three distinct notions in play. Real coefficient errors are measured by the ordinary real absolute value. Valuation truncation preserves coefficients *exactly* below an exponent bound.

Full-surreal order neighborhoods allow all surreal scales. Our numerical names combine the first two; they do not make the full surreal topology an effective metric space. Nor are they an admissible presentation of the pure valuation metric with the usual Cauchy representation on constants: nonzero real constants all have valuation zero, however small their real magnitudes.

3 The established structural notion

3.1 Hereditary effective cuts

Definition 3.1 (Structural presentation). A raw structural name is a program index e whose execution enumerates two collections of program indices, called its left and right options. Unfold these indices hereditarily into a rooted option tree. The name is *valid* if this tree is well founded and, at every node, the values of all left options are strictly below the values of all right options. Well-founded recursion assigns to the node the simplest surreal separating its options. The set $\mathcal{S}_{\text{str}}$ consists of the values of valid names.

An enumeration may never announce that it has finished. A leaf can be a program that runs forever without output. Thus validity is not the same as termination of the enumerating program. Repetition of an option has no semantic effect. A finite program describes a well-founded infinite object; no ordinary Turing machine is asserted to execute one step at every ordinal stage of its birthday.

The following result is an external classification, included to locate our work accurately rather than reproved here.

Theorem 3.2 (Known structural classification). *For hereditary effective surreal presentations, the values are exactly the surreals with hyperarithmetic sign sequences, interpreted on effective ordinal presentations. Equivalently, with the usual sign-sequence coding, they are $\mathbf{No} \cap L_{\omega_1^{\text{CK}}}$. This is a real-closed field. Its real slice is the hyperarithmetic reals, not merely $\mathbb{R}_c$.*

The effectivization program originates in [3, 4]; the displayed modern characterization and real-closedness statement are given in [5, 13, 6]. The full Lurie article was not accessible during preparation; its published abstract and metadata were checked, while the explicit characterization was checked against the later authors' public statements and lecture notes. Here hyperarithmetic means computable by effective transfinite recursion below the Church–Kleene ordinal; it is strictly broader than ordinary Turing computability. We use Theorem 3.2 only for the structural comparison, not in any proof of the numerical main theorem.

3.2 Elementary closure mechanisms

Proposition 3.3 (Structural ring compilation and computable cuts). *Negation, addition, and multiplication admit uniform compilers on valid structural names. If (x_n) and (y_n) are uniformly named families with $x_n < y_m$ for all n, m , then $\{\{x_n : n \in \mathbb{N}\} \mid \{y_n : n \in \mathbb{N}\}\}$ has a uniformly constructed valid structural name.*

Proof. For cuts, produce one root program which dovetails the two enumerations of names. Every branch below the root lies in a valid input tree; hence the new tree is well founded, and the promised inequalities give separation.

For arithmetic use the Conway option formulas. Negation interchanges sides and negates each option. Addition emits $x^L + y$ and $x + y^L$ on the left, and $x^R + y$ and $x + y^R$ on the right. Multiplication emits

$$x^L y + x y^L - x^L y^L, \quad x^R y + x y^R - x^R y^R$$

on the left, and the corresponding mixed left–right expressions on the right. These are finite syntactic expressions in recursively named smaller products. Standard effective self-reference supplies the indices of the recursive compilers. Well-founded induction on the input option trees, using the well-founded product order for pairs of operands, proves that the resulting finite-expression unfoldings are valid and have the intended values. Separation is exactly the classical Conway arithmetic theorem [1]. The construction never decides which of two arbitrary input values is larger. $\square$

This proof does not give a blind reciprocal compiler. The known assertion that $\mathcal{S}_{\text{str}}$ is a field must not be substituted for such an algorithm. In particular, restrictions to positive options in classical reciprocal formulas cannot simply be implemented by assuming a decidable sign test.

Proposition 3.4 (Computable ordinals and lack of an exhaustive enumeration). *Every computable ordinal belongs to $\mathcal{S}_{\text{str}}$. The set $\mathcal{S}_{\text{str}}$ is countable, but no uniform sequence of valid structural names is cofinal in it. In particular, no such sequence enumerates all its values.*

Proof. Given a computable well-order $(D, \prec)$, assign to each $d \in D$ the all-left name whose options are the names for $c \prec d$, and place all these names below a new all-left root. Induction identifies the root’s value with the order type of D . The compilers are effective since D and $\prec$ are decidable.

Countability follows from countably many program indices. If (x_n) is uniformly named, its all-left root has value $u = \{\{x_n : n \in \mathbb{N}\} \mid\}$ and satisfies $u > x_n$ for every n . This proves the cofinality assertion and the enumeration obstruction. $\square$

The same argument gives a positive lower bound $\{0 \mid \{p_n : n \in \mathbb{N}\}\}$ for every uniformly named family of positive values p_n . Neither operation is analogous to taking an ordinary real supremum or infimum.

3.3 A direct noncomputable real in the structural field

Proposition 3.5 (A halting real with a structural name). *Let K be the diagonal halting set. Then*

$$r_K = \sum_{e \geq 0} 2\chi_K(e)4^{-(e+1)}$$

is a noncomputable real with a valid structural name.

Proof. For each machine e , let a program initially enumerate no options. If machine e halts, it enumerates one left option with value zero. The resulting name h_e has value 0 or 1 according as $e \notin K$ or $e \in K$. The names h_e are uniform even though their values are not computable uniformly as real numbers.

Using the arithmetic compilers, construct names for $s_n = \sum_{e < n} 2h_e 4^{-(e+1)}$. Consider the structural cut

$$\{\{s_n - 4^{-n} : n \in \mathbb{N}\} \mid \{s_n + 4^{-n} : n \in \mathbb{N}\}\}. \quad (3.1)$$

Since $0 \leq r_K - s_n \leq \frac{2}{3} 4^{-n}$, every left bound is strictly below r_K and every right bound strictly above it. The shrinking real intervals determine every finite dyadic comparison with r_K . Their

simplest surreal separator is therefore the canonical real r_K : any other separator has all the finite sign information of r_K and is no simpler. This is also the standard embedding of a real by its rational cuts.

Finally, the base-four digits of r_K are only 0 and 2. Once the first e digits are known, the two possibilities for the next digit lie in separated intervals; the tail after that digit is at most one third of its nonzero contribution. Successively accurate Cauchy approximations would therefore decide all of K . Thus $r_K \notin \mathbb{R}_c$. $\square$

The general halting-real phenomenon is already present in Hamkins’s presentation [5]. The explicit margins in (3.1) are included to avoid the false assumption that a cut may use an option equal to its proposed value.

Corollary 3.6 (A structural–numerical incompatibility). *A notion which includes these uniform hereditary cut constructions and Conway arithmetic cannot also have exactly $\mathbb{R}_c$ as its real slice. In particular, there is no conversion of every real-valued structural name into an ordinary computable Cauchy name.*

Proof. Such a conversion would apply to (3.1) and compute r_K . $\square$

Proposition 3.7 (Birthday and ordinal boundary). *Every structurally computable surreal has birthday less than ω_1^{CK} . The ordinals belonging to $\mathcal{S}_{\text{str}}$ are exactly the computable ordinals.*

Proof. Unfold a valid enumerable dependency graph into a recursive tree of paths carrying finite witnesses for each enumerated edge. Its well-founded Kleene–Brouwer order is a computable well-order, so its rank is below ω_1^{CK} [7]. Induction bounds the birthday of each cut value by the rank of its construction. The birthday of an ordinal is that ordinal itself. Combine this bound with Proposition 3.4. $\square$

This is a necessary complexity bound, not a characterization by birthday: all ordinary real numbers have birthday at most ω , but not all are hyperarithmetic. Nor is a finite program carrying a semantic well-foundedness promise the same as an algorithm which can decide that promise.

For uniformly given Boolean-valued valid structural names b_n , the names $1 - b_n$ and $b_n b_m$ implement negation and conjunction, while $\{b_n - 1 : n \in \mathbb{N}\}$ implements countable disjunction. Its value is zero if every $b_n = 0$, and one otherwise. The empty right side matters: putting 1 on that side would give 1/2 in the true case. This elementary mechanism helps explain the imported hyperarithmetic classification without purporting to reprove it.

3.4 Validity is a genuinely higher-order promise

Proposition 3.8 (Validity has full well-foundedness complexity). *For the program coding in Definition 3.1, the set of valid root indices is Π_1^1 -complete under computable many-one reductions.*

Proof. For hardness, start with a recursive tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$. At a node $\sigma \in T$, put the programs for its immediate successors on the left and nothing on the right. If T is well founded, every cut is automatically separated and the root is valid. If T has an infinite branch, the dependency graph is not well founded. The reduction is effective. Well-foundedness of recursive trees is Π_1^1 -complete [7].

For the upper bound, well-foundedness of the reachable dependency graph is Π_1^1 . On a well-founded graph, the recursive comparison relation on pairs of positions is uniquely determined by

$$a \preceq b \iff (\forall c \in L_a \neg(b \preceq c)) \wedge (\forall d \in R_b \neg(d \preceq a)). \quad (3.2)$$

This recursion is well founded on pairs of ranks. Existence and uniqueness hold even before knowing that every position is a number: it is the usual comparison recursion for well-founded games. The assertion that a set of pairs satisfies (3.2) is arithmetical in that set and the program index. Hereditary separation is then the arithmetical requirement that, at every reachable node, every left option is strictly below every right option according to this relation.

Thus, in addition to well-foundedness, require that *every* set of pairs satisfying the recursive comparison equations satisfies hereditary separation. This is a universal second-order quantification over an arithmetical predicate. On a well-founded graph the unique comparison relation exists, so the assertion is equivalent to validity and is not vacuous. This gives a Π_1^1 definition. $\square$

In particular there is no algorithm that recognizes exactly the valid hereditary programs and halts on all of them. A proof-carrying subclass can be effectively checked in a chosen proof system, but cannot exhaust the semantic validity predicate.

4 A computable order presentation can hide a noncomputable real

A tempting alternative is to call a surreal computable when its signs are a computable labeling of a computably presented well-order. The phrase sounds like direct sign access, but it need not provide access to the sign at the n th ordinal position. A computable presentation of an order of type ω need not have a computable order isomorphism with the standard $\mathbb{N}$.

Theorem 4.1 (The signed-order presentation trap). *For every computably enumerable set $A \subseteq \mathbb{N}$, there is a decidable domain $D_A \subseteq \mathbb{N}^2$, a decidable order on D_A of order type ω , and a computable labeling by $+$ and $-$, such that the resulting canonical sign sequence denotes a nondyadic real r_A of Turing degree exactly that of A .*

Proof. Choose a computable stage enumeration in which an element enters A at most once. For every e , put $(e, 0)$ and $(e, 1)$ in D_A . Also put $(e, s + 2)$ in D_A exactly when e first enters A at stage s . Membership is decidable by simulating the enumeration through the specified finite stage. Order D_A lexicographically. Each e contributes a nonempty finite block of size two or three, so the order type is ω .

Label $(e, 0)$ by $+$ and every other point by $-$. The sign sequence in ordinal order is the concatenation

$$+ \ -^{1+\chi_A(0)} \ + \ -^{1+\chi_A(1)} \ + \ -^{1+\chi_A(2)} \ \dots \quad (4.1)$$

Both signs occur infinitely often. Its first two signs are $+-$, and its successive dyadic intervals shrink to a nondyadic real in $(0, 1)$. With oracle A , one computes the blocks and hence obtains an ordinary fast real name. Therefore $r_A \leq_T A$.

Conversely, a fast name for a nondyadic real computes any prescribed finite initial segment of its surreal sign sequence. At each step compare with the relevant dyadic pivot. Equality never occurs, and eventually the approximation interval decides the strict comparison. After recovering sufficiently many signs of (4.1), count the one or two minus signs following the e th plus sign. This recovers $\chi_A(e)$. Hence $A \leq_T r_A$. $\square$

The theorem uses a particularly tame order: all blocks have size at most three. What fails is not decidability of comparison or boundedness of block size, but effective identification of the successor after a possibly missing point. Thus one cannot establish real compatibility merely by placing computability requirements on an abstract order and its labels.

For ordinary reals, *canonical* sign access and Cauchy computation agree at the level of computable values, with the familiar nonuniform treatment of terminating dyadic expansions. For transfinite sign sequences, a proposed representation must also specify ordinal addresses, restrictions to initial segments, and the navigation information needed by its algorithms. The coefficient representation developed next avoids this address problem by supplying explicit rational coefficient addresses and effective finite search data.

5 Why computable Hahn coefficients are insufficient

Suppose we ask only that $q \mapsto a_q$ be computable, and that its nonzero support be well ordered. Mathematical Hahn multiplication is defined because each coefficient receives only finitely many contributions. It does not follow that one can find those contributions or know that the search for them is complete.

The next construction strengthens this objection. The supports are not merely well ordered: they are decidable, left-finite, and have computably bounded numbers of points below every exponent threshold.

Theorem 5.1 (Decidable supports with a noncomputable product). *There are two rational-exponent Hahn series A, B , each with coefficients in $\{0, 1\}$ and a computable exact coefficient function on $\mathbb{Q}$, whose supports are decidable and left-finite, but whose product does not have a uniformly Cauchy-computable coefficient function.*

Proof. Let $T(e)$ be the first halting stage when $e \in K$. Put

$$c_e = 4^{e+1}, \quad \delta_s = \frac{1}{s+2},$$

and define

$$A = \sum_{e \in K} t^{c_e + \delta_{T(e)}}, \quad B = \sum_{e \in K} t^{2c_e - \delta_{T(e)}}. \quad (5.1)$$

Each support has at most one point in its e th block. The A blocks lie in $(c_e, c_e + \frac{1}{2}]$ and the B blocks in $[2c_e - \frac{1}{2}, 2c_e)$. They are separated and tend to infinity, proving left-finiteness and a computable bound on the number of points below any rational threshold.

For exact coefficient access to A , given $q \in \mathbb{Q}$, determine whether it lies in one of the finitely searchable intervals $(c_e, c_e + \frac{1}{2}]$. If it does, compute $s = 1/(q - c_e) - 2$. A nonnegative integral value s is necessary; it is then sufficient to simulate machine e for s stages and ask whether its first halt was exactly at that stage. All other cases give coefficient zero. The same algorithm for B uses $s = 1/(2c_e - q) - 2$. This proves computability of the coefficient functions and decidability of the supports.

We claim

$$[t^{3c_e}](AB) = \chi_K(e). \quad (5.2)$$

A contributing pair indexed by $i, j \in K$ must satisfy

$$c_i + 2c_j - 3c_e = \delta_{T(j)} - \delta_{T(i)}.$$

The left side is an integer divisible by four; the right side has absolute value less than $1/2$. Hence both sides vanish. Uniqueness of finite base-four expansions in

$$4^{i+1} + 2 \cdot 4^{j+1} = 3 \cdot 4^{e+1}$$

forces $i = j = e$: distinct indices would produce a digit 1 and a digit 2 in different positions, rather than a single digit 3. When $e \in K$, that one pair is present and its perturbations cancel. Otherwise no pair exists. This proves (5.2).

A joint Cauchy algorithm for the product coefficients, requested to error less than $1/4$ at $3c_e$, would distinguish zero from one and decide K . The product is nevertheless an ordinary well-defined left-finite Hahn series. Its failure is effective, not algebraic. $\square$

The obstruction is not the impossibility of comparing real coefficients: all input coefficients are exact integers. Nor is it the absence of a cardinality bound on the finite work. The missing information is a *complete finite list of candidate exponents*. A bound saying “at most one point in this block” does not reveal whether the point exists or which rational encodes its halting time.

Corollary 5.2. *The class of left-finite Hahn series with uniformly computable coefficient functions is not, by itself, a field or even a ring under Hahn multiplication. Decidable support membership and computable bounds on finite-section cardinalities do not repair this failure.*

This motivates a stronger numerical representation, but one which still avoids requiring an impossible exact zero test on real coefficients.

5.1 A complementary obstruction with singleton fibers

The preceding construction keeps *both* supports left-finite. The next, retained from C, instead has a global bound of one on every convolution fiber; one factor has a bounded accumulation point. These are distinct strengthenings of the same negative principle, so neither is discarded.

Theorem 5.3 (Sparse Hahn multiplication can compute the halting set). *There are Hahn series F, G with computable coefficient functions taking values in $\{0, 1\}$ and decidable well-ordered supports, such that FG has no computable coefficient function. Every convolution fiber for these two supports has size at most one.*

Proof. For a halting machine e , write $s_e \geq 0$ for its exact halting stage. Define

$$F = \sum_{n \geq 0} t^{1-1/(n+2)}, \quad G = \sum_{e \in K} t^{e+1/(s_e+2)}. \quad (5.3)$$

The support of F is strictly increasing, of order type ω . Its coefficient at $q = 1$ is zero; at any other rational q , test whether $(1 - q)^{-1}$ is an integer at least two. The support of G contains at most one element in each interval $(e, e + 1/2]$, so is also well ordered. For a rational query q , let $e = \lfloor q \rfloor$. Test whether $e \geq 0$ and $q - e = 1/(s + 2)$ for an integer $s \geq 0$. If so, simulate machine e through stage s and test whether this is its exact halting stage. This decides the coefficient of G . Thus both coefficient functions and both supports are decidable.

An exponent in the product has the form

$$j + 1 + \frac{1}{s_j + 2} - \frac{1}{n + 2}.$$

For fixed target exponent q , the integer j must lie in $(q - 3/2, q - 1/2)$, since $1 - 1/(n + 2) \in [1/2, 1)$ and $1/(s_j + 2) \in (0, 1/2]$. This open interval of length one contains at most one integer. For each j there is at most one support term of G , and then at most one possible n . Thus every fiber has at most one member. The ordinary Hahn product is well defined.

At the integer target $q = e + 1$, the only possible integer block is $j = e$: the difference $1/(s_j + 2) - 1/(n + 2)$ is strictly between $-1/2$ and $1/2$, so it cannot compensate a nonzero integer. The target is attained exactly when e halts and $n = s_e$. Consequently

$$[t^{e+1}](FG) = \chi_K(e). \quad (5.4)$$

A uniform approximation algorithm for the coefficients of FG , requested at these targets with error at most $1/4$, would decide K . Therefore FG has no computable coefficient function. $\square$

The statement is stronger than nonexistence of a uniform multiplication algorithm: the product of these two fixed computable coefficient functions is not a computable coefficient function at all. Each individual coefficient in (5.4) is of course a computable rational. It is their uniform recovery that fails.

Classical support lemmas establish that each convolution coefficient is a finite sum. They do not produce an effective stopping criterion for locating its summands. Even a known global bound of one on their number does not produce such a criterion. This is the same general danger as replacing an unknown finite search domain by a promise that only finitely many answers exist.

B uses the equivalent collision template $\alpha_s = 1 - 2^{-s-1}$, $F_B = \sum_s t^{\alpha_s}$, and $G_B = \sum_{e \in K} t^{2e+2-\alpha_{s_e}}$. At target $2e+2$, equality of the two fractional offsets forces the same halting stage, and the coefficient is $\chi_K(e)$. Rational queries recover and check only a finite candidate stage. The singleton-fiber formulation above and the left-finite formulation preceding it retain the stronger conclusions; the dyadic offsets are an alternative fixture, not a conflicting definition of multiplication.

5.2 The sharp oracle cost of an unrestricted convolution

Theorem 5.4 (One jump for coefficients, not for an ordinary output name). *Let two Hahn series with rational exponents have uniformly computable real coefficient functions and well-ordered supports, but no effective support covers. Their product coefficient function is uniformly computable in $0'$. The bound is sharp, already for exact rational zero-one coefficients.*

Proof. For a fixed rational target q , the set

$$E_q = \{r \in \mathbb{Q} : a_r b_{q-r} \neq 0\}$$

is computably enumerable, since a nonzero computable real is semidecidable. It is finite: infinitely many such r would yield a strictly increasing sequence in the first well-ordered support and hence a strictly decreasing sequence $q - r$ in the second. This contradicts well-ordering.

Keep a finite list D of contributors already detected. A halting oracle decides whether the semidecision search for another member of $E_q \setminus D$ will succeed. If it will, run that search until a witness appears and add it; otherwise stop. Finiteness ensures termination with exactly E_q . Compute the finite sum of the named real products to any requested precision. For exact rational coefficients this produces an exact rational result. Either counterexample above gives product coefficients which decide K , so any oracle uniformly solving this task computes $0'$. $\square$

The witness search need not find a *least* rational contributor and does not require a decidable support test. This formulation also covers Cauchy-real coefficients correctly. The resulting coefficient algorithm may need $0'$ anew at every target q ; in general no ordinary computable product name exists. This differs from the finite normalization advice in Theorem 12.3, after which an ordinary inverse program is returned. The theorem supplies coefficients, not an effective finite-cover name for the product. Structured positive results for special Hahn families, such as Mahler-equation support receptacles [17], impose additional data and do not contradict this obstruction.

6 Effective left-finite Cauchy–Hahn names

Definition 6.1 (The numerical representation). An *effective left-finite Cauchy–Hahn name* consists of a rational number ℓ and two total computable procedures A, F with the following meaning. There is a real coefficient family $(a_q)_{q \in \mathbb{Q}}$ such that

- (1) $a_q = 0$ whenever $q < \ell$;
- (2) $A(q, k) \in \mathbb{Q}$ and $|A(q, k) - a_q| \leq 2^{-k}$ for every q, k ;
- (3) for each $B \in \mathbb{Q}$, $F(B)$ is a finite rational set contained in $[\ell, B)$ and containing every $q < B$ with $a_q \neq 0$.

Its value is $x = \sum_q a_q t^q$, viewed in **No** through normal forms. The set $\mathcal{L}_c$ consists of the values of such computable names.

The procedure F is a *finite-section cover*, not an exact support oracle. It may list exponents with coefficient zero, and no test for deleting such entries is required. Lists can always be converted effectively into finite sets of reduced rational numbers. They need not be nested as B increases. For $B \leq \ell$ the only allowed list is empty.

These small details matter. Requiring an exact nonzero-support list would already demand a zero test for a single arbitrary computable real constant. Allowing harmless zero candidates makes addition and cancellation effective. The name’s correctness conditions remain promises; the algorithms below preserve them rather than attempting to decide them.

Proposition 6.2 (Well-definedness and the real slice). *Every valid numerical name defines a unique left-finite Hahn series. The set $\mathcal{L}_c$ is countable, contains every rational monomial t^q , and satisfies $\mathcal{L}_c \cap \mathbb{R} = \mathbb{R}_c$. The inclusion of ordinary computable real constants into this representation is uniform.*

Proof. The error bounds determine each real a_q uniquely. The covers imply left-finiteness; hence the series is a valid Hahn normal form. Its surreal value is unique by the normal-form theorem. Countably many programs and finite rational data give countably many names.

For t^q , use lower bound q , coefficient one at q and zero elsewhere, and cover $\{q\}$ exactly when $B > q$. For a Cauchy-named real a , use lower bound zero, coefficient a at exponent zero and zero elsewhere, and cover $\{0\}$ when $B > 0$. This construction is uniform even when $a = 0$. Conversely, if the value of a name is an ordinary real, uniqueness of normal form says that its only possible nonzero coefficient is a_0 . The algorithm $A(0, k)$ therefore computes that real. $\square$

Definition 6.3 (Finite jet). For $B \in \mathbb{Q}$ define the exact finite jet

$$J_B x = \sum_{q \in F(B)} a_q t^q.$$

Its data consist of a finite rational exponent list and Cauchy names for its real coefficients, not finite rational approximations to those coefficients.

Proposition 6.4 (Certified truncation). *From a numerical name and B one computes a name for $J_B x$, and*

$$v(x - J_B x) \geq B.$$

The value of $J_B x$ is independent of the particular valid cover.

Proof. All nonzero coefficients below B occur in the list, and the remaining listed coefficients are zero. Thus the coefficients of the difference below B vanish. Restrict the original coefficient oracle to the finite list to obtain a name for the jet. $\square$

The series in Theorem 5.1 have no computable covers of this kind. For example, a cover of A below $c_e + 1$ would give finitely many candidates in its e th block. Its exact coefficient oracle could inspect them and decide whether $e \in K$.

7 Uniform ring operations and certified reciprocation

7.1 Finite Cauchy arithmetic

We repeatedly use the elementary fact that finite sums and products of Cauchy-computable real numbers are computable uniformly. This does not require zero tests. A precision-zero approximation gives a computable bound $|a| \leq |A(0)| + 1$. To approximate a finite sum of products, choose a common input error $\eta \leq 1$ small enough that the sum of the estimates

$$|ab - \tilde{a}\tilde{b}| \leq \eta(|a| + |b| + 1)$$

is below the requested output error. All required bounds follow from coarse approximations. If $a \neq 0$ is promised, its reciprocal is computable as an ordinary real: search until an approximation interval excludes zero and then propagate errors through division using the resulting positive lower bound for $|a|$.

Theorem 7.1 (Uniform numerical ring arithmetic). *Addition, negation, and multiplication are computable uniformly on numerical names. For rational r , the operation $x \mapsto t^r x$ is also uniformly computable.*

Proof. Negation keeps the lower bound and cover. For addition use lower bound $\min(\ell_x, \ell_y)$, coefficient sums, and the cover $F_x(B) \cup F_y(B)$. Rational shifting adds r to the lower bound and exponents, uses a_{q-r} as coefficient, and shifts $F_x(B - r)$ by r .

For multiplication put $\ell = \ell_x + \ell_y$. At an exponent q the exact coefficient is the finite sum

$$[t^q](xy) = \sum_{r \in F_x(q - \ell_y + 1)} ([t^r]x)([t^{q-r}]y). \quad (7.1)$$

Indeed, a nonzero contribution requires $r \geq \ell_x$ and $q - r \geq \ell_y$, so $r \leq q - \ell_y$ and lies in the indicated cover. Zero candidates contribute nothing. Finite Cauchy arithmetic computes this coefficient to any requested precision.

A support cover for the product below B is

$$F_{xy}(B) = (F_x(B - \ell_y) + F_y(B - \ell_x)) \cap [\ell, B), \quad (7.2)$$

where the plus sign between finite sets denotes their pairwise sum set. If $r + s < B$ contributes to a nonzero coefficient, then $r < B - \ell_y$ and $s < B - \ell_x$. Thus (7.2) contains every possible nonzero exponent below B . It is a computable finite set. This proves all requirements on the product name. $\square$

No test for cancellation was hidden in either formula. In particular, the output may have a much larger true leading exponent than its supplied lower bound, or may be zero.

7.2 An effective positive gap

Lemma 7.2 (Gap extraction without normalization). *Suppose a numerical name is promised to satisfy $\text{supp}(h) \subseteq \mathbb{Q}_{>0}$. One can compute $\delta \in \mathbb{Q}_{>0}$ such that $v(h) \geq \delta$, including the case $h = 0$.*

Proof. Compute $F_h(1)$, discard its nonpositive candidates, and take

$$\delta = \min(\{1\} \cup (F_h(1) \cap \mathbb{Q}_{>0})).$$

Every nonzero exponent of h below one occurs among these positive candidates; every other nonzero exponent is at least one. The promise justifies discarding nonpositive candidates, even though their zero coefficients were not algorithmically recognized. $\square$

Under the same promise we may replace the lower bound by zero and discard all nonpositive candidates in every cover. This is *certified filtering*: a theorem about the input, rather than an equality decision, authorizes removal of those positions.

Lemma 7.3 (Effective geometric series). *For a numerically named h with positive support, the series $G(h) = \sum_{n \geq 0} h^n$ has a uniformly computable numerical name. It satisfies $(1 - h)G(h) = 1$ and, for every $N \geq 0$,*

$$(1 - h) \sum_{n=0}^N h^n = 1 - h^{N+1}.$$

Proof. Extract a positive gap δ . Each h^n is numerically computable by Theorem 7.1 and has support in $[n\delta, \infty)$. At any threshold B , only the finitely many n with $n\delta < B$ can contribute. The union of their computed covers is a finite cover for $G(h)$ below B . For a queried exponent q , choose N with $(N + 1)\delta > q$ and compute the coefficient of the finite sum through N . This stabilizes the exact coefficient, after which its real precision is computed separately. A lower bound is zero.

Finite multiplication proves the displayed remainder identity. At every fixed coefficient, h^{N+1} eventually contributes zero. Hence the coefficientwise Hahn identity $(1 - h)G(h) = 1$ follows. This reasoning takes place in the fixed left-finite workspace, not in the full surreal topology. $\square$

Theorem 7.4 (Field closure and certified inversion). *Every nonzero $x \in \mathcal{L}_c$ has $x^{-1} \in \mathcal{L}_c$. Uniform inversion is possible from a numerical name supplemented by its true leading exponent $\lambda = v(x) \in \mathbb{Q}$.*

Proof. The coefficient $a = [t^\lambda]x$ is a nonzero computable real. With the nonzero promise compute its real inverse. Form

$$h = 1 - a^{-1}t^{-\lambda}x.$$

The supplied leading-exponent certificate proves that all coefficients of h at nonpositive exponents vanish. Lemma 7.3 now gives

$$x^{-1} = a^{-1}t^{-\lambda} \sum_{n \geq 0} h^n \quad (7.3)$$

uniformly, with lower bound $-\lambda$ and covers obtained by shifting those of the geometric series. The exact identity follows from $1 - h = a^{-1}t^{-\lambda}x$.

For extensional closure, every particular nonzero left-finite series has a least support exponent, which is a rational number. That single rational can be hard-coded into an inverse program. This proves existence of a computable inverse name without claiming a uniform procedure for finding that rational from an arbitrary input name. $\square$

This final distinction is the source of the apparent paradox: all inverse values are computable in the specified sense, but one cannot uniformly choose their names from all nonzero input names. The obstruction will be proved, rather than merely asserted, in Section 12.

7.3 The valuation ring and standard part

Define

$$\mathcal{O} = \{x \in \mathcal{L}_c : v(x) \geq 0\}, \quad \mathfrak{m} = \{x \in \mathcal{L}_c : v(x) > 0\}.$$

These include zero by the convention $v(0) = +\infty$.

Proposition 7.5 (Value group and effective residue). *The value group of $\mathcal{L}_c$ is $\mathbb{Q}$. The ring $\mathcal{O}$ is convex in its ordered field and has maximal ideal $\mathfrak{m}$. Coefficient extraction gives*

$$\text{st} : \mathcal{O} \longrightarrow \mathbb{R}_c, \quad \text{st}(x) = [t^0]x, \quad \mathcal{O}/\mathfrak{m} \cong \mathbb{R}_c.$$

This standard-part operation is computable from a numerical name with the promise $x \in \mathcal{O}$, without knowing $v(x)$.

Proof. Every nonzero valuation is a rational exponent, and $v(t^q) = q$ shows surjectivity onto $\mathbb{Q}$. An element has nonnegative valuation exactly when it is finite as a surreal: a negative leading exponent yields magnitude larger than every positive integer, and a nonnegative exponent gives a real constant plus an infinitesimal. Thus $\mathcal{O}$ is a convex subring.

For two elements of $\mathcal{O}$, only their exponent-zero terms contribute to the exponent-zero coefficient of their product. Hence st is a ring homomorphism. Its kernel is $\mathfrak{m}$, and constants give surjectivity. The quotient is a field, proving maximality. Finally, $A(0, k)$ is already an ordinary Cauchy name for the residue. $\square$

Each finite element has the unique decomposition $x = \text{st}(x) + (x - \text{st}(x))$ with infinitesimal second summand. The algorithm extracts it without deciding whether that infinitesimal summand is zero.

8 Effective Hensel lifting and real closedness

8.1 A quantitative simple-root algorithm

For a polynomial $P \in \mathcal{O}[Z]$, denote by $\overline{P} \in \mathbb{R}_c[Z]$ its coefficientwise residue. A residue root c is simple if $\overline{P}(c) = 0$ and $\overline{P}'(c) \neq 0$.

Theorem 8.1 (Effective simple Hensel lifting). *Let a finite-degree polynomial $P \in \mathcal{O}[Z]$ be given by numerical names for its coefficients, and let $c \in \mathbb{R}_c$ be given by a Cauchy name. Under the promises*

$$\overline{P}(c) = 0, \quad a = \overline{P}'(c) \neq 0,$$

one can compute a numerical name for the unique $y \in \mathcal{O}$ satisfying $P(y) = 0$ and $y - c \in \mathfrak{m}$. More precisely, one computes a positive rational δ and numerical Newton iterates y_n such that

$$v(P(y_n)) \geq 2^n \delta, \quad v(y - y_n) \geq 2^n \delta. \quad (8.1)$$

Proof. Put $y_0 = c$. Since $P(c)$ has positive support, the gap lemma gives $\delta > 0$ with $v(P(c)) \geq \delta$. This also works if $P(c) = 0$. Inductively define

$$y_{n+1} = y_n - P(y_n)P'(y_n)^{-1}. \quad (8.2)$$

If $y_n \equiv c \pmod{\mathfrak{m}}$, then $P'(y_n)$ has residue $a \neq 0$ and hence *known* leading exponent zero. Its inverse is therefore computed by the certified algorithm, without a general normalization procedure. The correction has positive valuation, so $y_{n+1} \equiv c \pmod{\mathfrak{m}}$.

All polynomial coefficients and iterates lie in $\mathcal{O}$. The finite Taylor identity, with $u = -P(y_n)/P'(y_n)$, gives

$$P(y_n + u) = P(y_n) + P'(y_n)u + u^2 R_n, \quad R_n \in \mathcal{O}.$$

The first two terms cancel. If $v(P(y_n)) \geq 2^n \delta$, the unit $P'(y_n)$ has valuation zero, so $v(u) \geq 2^n \delta$ and $v(P(y_{n+1})) \geq 2^{n+1} \delta$. Induction proves these estimates and $v(y_{n+1} - y_n) \geq 2^n \delta$.

For each $q \in \mathbb{Q}$, choose n with $2^n \delta > \max(q, 0)$ and define the coefficient of y at q to be that of y_n . All later iterates agree at that exponent, so this is consistent. It gives a uniform Cauchy coefficient algorithm. Set every coefficient at a negative exponent to zero, as justified by the invariant $y_n \in \mathcal{O}$. For a threshold $B > 0$, choose n with $2^n \delta > B$ and use $F_{y_n}(B) \cap [0, B)$ as a cover for y ; for $B \leq 0$, use the empty set. Thus y has a numerical name with lower bound zero, and the second inequality in (8.1) follows.

For any fixed bound, polynomial evaluation uses only finitely many earlier coefficients, since all arguments and coefficients have nonnegative valuation. Consequently $P(y)$ agrees below that bound with $P(y_n)$ for sufficiently large n , and the first inequality in (8.1) makes those coefficients zero. Thus $P(y) = 0$.

For uniqueness, if $z \equiv c \pmod{\mathfrak{m}}$ is another root, the polynomial difference quotient gives

$$P(y) - P(z) = (y - z)U, \quad U \equiv P'(c) \equiv a \pmod{\mathfrak{m}}.$$

The factor U is a unit, so $y = z$. □

The number of Newton rounds needed to determine an exponent jet grows logarithmically in B/δ . This is a bound on rounds, not a polynomial bit-complexity claim: cover sizes, coefficient magnitudes, and real precision costs may grow substantially. The theorem is effective only under its promises; it does not claim to decide whether a proposed residue root is simple.

8.2 Why the computable reals are real closed

For completeness, the extensional real-algebraic fact needed below has an elementary proof which does not require a uniform multiplicity decision.

Lemma 8.2. *Every real root of a nonzero polynomial with coefficients in $\mathbb{R}_c$ belongs to $\mathbb{R}_c$. Consequently $\mathbb{R}_c$ is real closed.*

Proof. Let r be a real root of multiplicity m of P . The polynomial $Q = P^{(m-1)}$ has a simple root at r . Choose rational endpoints $u < r < v$ and a positive rational d such that Q' has constant sign and $|Q'(x)| > d$ throughout $[u, v]$, and r is the unique root there. These finitely many data exist by continuity and simplicity and may be hard-coded for this particular root.

For a requested precision k , search through rational $q \in (u, v)$ and increasing computation precisions until $|Q(q)| < d 2^{-k}$ is certified. Polynomial evaluation at a rational is Cauchy computable, and this strict inequality is semidecidable. Rationals arbitrarily close to r ensure that the search succeeds. The mean value theorem then gives $|q - r| < 2^{-k}$. Hence r is computable.

Every real element algebraic over $\mathbb{R}_c$ is such a root. Since $\mathbb{R}$ is real closed, positive square roots and roots of odd-degree polynomials exist there and therefore already lie in $\mathbb{R}_c$. These two properties characterize real-closed ordered fields. $\square$

The proof chooses m, u, v, d nonuniformly. This is sufficient for the algebraic statement and should not be mistaken for a root-isolation algorithm that determines multiplicities from arbitrary coefficient names.

8.3 The one external valuation theorem

We use the following standard consequence of the lemma of Ostrowski; the formula and the residue-characteristic-zero specialization are recorded explicitly in [9, §2.1, equation (1)]:

$$[E : F] = (\mathbf{v} E : \mathbf{v} F) [Ev : Fv] \quad (8.3)$$

for a finite extension of a henselian valued field F of residue characteristic zero, with its unique extended valuation. Here Fv and Ev denote residue fields. The general formula has an additional defect factor, a power of the residue characteristic exponent; in characteristic zero that exponent is one. Henselianity, equivalently uniqueness of the valuation extension, is essential.

Lemma 8.3 (A real-closedness criterion). *Let F be an ordered field with a convex henselian valuation. If its value group is divisible and its residue field is real closed of characteristic zero, then F is real closed.*

Proof. Let E/F be a finite ordered algebraic extension. The convex hull in E of the valuation ring of F is a convex valuation ring extending it. Indeed, a convex subring of an ordered field is a valuation ring: if x does not lie in it, $|x|$ exceeds every positive element of the ring, so $|x^{-1}| < 1$ and x^{-1} lies in it. Its intersection with F is the original convex ring. The induced residue field is ordered, extending the order on Fv . Henselianity makes this the unique extended valuation.

The fundamental inequality makes both the value-group index and residue degree finite. A torsion-free ordered group cannot properly contain a divisible subgroup with finite index: for $\gamma \in \mathbf{v} E$, some $n\gamma$ lies in $\mathbf{v} F$; divisibility supplies $\eta \in \mathbf{v} F$ with $n\eta = n\gamma$, and torsion-freeness gives

$\eta = \gamma$. Thus the ramification index is one. The finite ordered algebraic extension Ev/Fv is trivial because Fv is real closed, so the residue degree is one. Equation (8.3) gives $[E : F] = 1$.

If F were not real closed it would have a proper ordered algebraic extension, containing a proper finite subextension generated by one element. The preceding argument rules this out. $\square$

Theorem 8.4 (Real closedness of the effective core). *The ordered field $\mathcal{L}_c$ is real closed, and $\mathcal{L}_c[i]$ is algebraically closed.*

Proof. Field closure was proved in Theorem 7.4. The valuation is convex, has value group $\mathbb{Q}$, and has residue field $\mathbb{R}_c$, by Proposition 7.5. Theorem 8.1 gives the simple root form of Hensel's lemma, hence henselianity. The group $\mathbb{Q}$ is divisible, and Lemma 8.2 makes the characteristic-zero residue field real closed. Apply Lemma 8.3. The usual algebraic characterization of a real-closed field says that adjoining a square root of -1 gives its algebraic closure. $\square$

This proves more than the existence of square roots of infinitesimal perturbations of one: *every* odd-degree polynomial over the effective core has a root in that core, regardless of the scales or multiplicities of its roots. The general existence assertion is extensional. The uniform root algorithms proved here require stated residue data or a finite-jet branch certificate; neither is a blind branch selector.

Corollary 8.5 (Algebraic operations on individual computable surreals). *Every real surreal root of a nonzero polynomial over $\mathcal{L}_c$ belongs to $\mathcal{L}_c$. Every surcomplex root of such a polynomial belongs to $\mathcal{L}_c[i]$.*

Proof. A real-closed ordered subfield has no proper ordered algebraic extension. Any real surreal root generates such an extension unless it already lies in the field. For a surcomplex root, the polynomial already splits over $\mathcal{L}_c[i]$, which is algebraically closed, so every root in a larger field is one of its roots there. $\square$

9 The computable Puiseux subfield

A particularly transparent subfield uses a fixed denominator for all exponents of each element, although different elements may have different denominators.

Definition 9.1 (Computable Puiseux series). Let $\mathcal{P}_c$ consist of all series

$$x = \sum_{n \geq N} a_n t^{n/m}, \quad m \in \mathbb{N}_{>0}, \quad N \in \mathbb{Z}, \quad (9.1)$$

with uniformly Cauchy-computable coefficients $a_n \in \mathbb{R}_c$. A name consists of m, N and this joint coefficient algorithm.

Proposition 9.2 (Embedding and arithmetic). *Puiseux names translate uniformly to numerical names. The set $\mathcal{P}_c$ is a subfield of $\mathcal{L}_c$. Addition and multiplication are uniform; inversion is uniform when the leading integer index is supplied.*

Proof. Use lower bound N/m . At a rational exponent q , return a_{mq} if mq is an integer at least N , and zero otherwise. A cover below B is the finite grid $\{n/m : N \leq n < mB\}$. For two series pass to

denominator $\text{lcm}(m_1, m_2)$, inserting zero coefficients at the other grid points. Finite convolution computes sums and products.

For a nonzero series with leading index ν , write $x = t^{\nu/m} \sum_{j \geq 0} a_{\nu+j} t^{j/m}$. Its inverse coefficients are computed successively from

$$b_0 = a_\nu^{-1}, \quad b_n = -a_\nu^{-1} \sum_{j=1}^n a_{\nu+j} b_{n-j} \quad (n \geq 1), \quad (9.2)$$

so $x^{-1} = t^{-\nu/m} \sum_{n \geq 0} b_n t^{n/m}$. Each requested coefficient requires only a finite recursive computation with real Cauchy arithmetic. For an individual nonzero value, its integer ν can be hard-coded. $\square$

Theorem 9.3 (A proper real-closed subfield). *The field $\mathcal{P}_c$ is real closed and is a proper subfield of $\mathcal{L}_c$.*

Proof. For a finite polynomial over the valuation ring of $\mathcal{P}_c$, choose a common exponent denominator m for its coefficients. In the simple Hensel construction, all Newton iterates and their inverses stay on this same $\frac{1}{m}\mathbb{Z}$ grid. The stabilized coefficients are uniformly computable, with lower index zero for the lifted integral root. Thus the proof of Theorem 8.1 gives henselianity inside $\mathcal{P}_c$, not merely inside $\mathcal{L}_c$. Its value group is still $\mathbb{Q}$, since it contains every t^q , and its residue field is $\mathbb{R}_c$. Lemma 8.3 proves real closedness.

For strictness consider

$$u = \sum_{n \geq 1} t^{n+1/(n+1)}. \quad (9.3)$$

The exponent sequence is computable, strictly increasing, and unbounded. A coefficient query q only requires testing the finitely many integers $1 \leq n < q$ for the exact equation $q = n + 1/(n + 1)$. A cover below B is computed by checking $1 \leq n < B$. This gives a numerical name for u . But the reduced denominator of $n + 1/(n + 1)$ is exactly $n + 1$, since $\gcd(n(n + 1) + 1, n + 1) = 1$. These denominators cannot all divide any fixed integer m . Uniqueness of Hahn normal form shows $u \notin \mathcal{P}_c$. $\square$

No Newton–Puiseux algorithm was silently assumed here. Real closedness follows from the effective unit-root iteration plus an explicit standard valuation criterion. General roots may require ramification or nonuniform branch data; the theorem does not compute all those data blindly.

Example 9.4 (Explicit computable infinitesimals). All of

$$\frac{1}{1-t} = \sum_{n \geq 0} t^n, \quad e^t = \sum_{n \geq 0} \frac{t^n}{n!}, \quad \sum_{n \geq 0} \pi^{n^2} t^{n/3}$$

belong to $\mathcal{P}_c$. There is no real-analytic growth condition on the coefficients of a formal infinitesimal series: the last example is allowed regardless of the radius of convergence of a related ordinary real power series. Its grid supplies finite exponent sections.

The numerical core is deliberately not closed under all small Conway cuts. Every element of $\mathcal{L}_c$ is eventually dominated in magnitude by some ω^n , because its leading exponent is rational. Thus the uniformly named sequence $1, \omega, \omega^2, \dots$ is cofinal in its positive magnitudes. Its all-left Conway cut is ω^ω , which cannot lie in $\mathcal{L}_c$: its normal form has exponent $-\omega$ in the t convention, not a rational exponent. Structural computability accommodates this cut; our fixed rank-one numerical workspace does not. Larger scales require larger workspaces, not a change in the meaning of rational exponents.

10 Finite jets and real-closedness

The arguments in B and C are the same finite-advice mechanism, written in slightly different normalizations. We retain C's explicit divided-derivative formulation and B's emphasis on a single fixed coefficient inverse. It is not enough to invoke Newton–Puisseux and observe that every root coefficient is a computable real. The coefficients of one root must have a *single* algorithm. This section proves that uniformity. The proof is local at a simple root and needs only a finite amount of nonuniform seed data.

For any coefficient field k , write $\mathcal{P}(k) = \bigcup_{d \geq 1} k((t^{1/d}))$ without an effectiveness restriction. Importantly, $\mathcal{P}_c \subsetneq \mathcal{P}(\mathbb{R}_c)$: the series $\sum_{n \geq 0} \chi_K(n)t^n$ has individually computable rational coefficients but no jointly computable coefficient function. In every Puiseux name, coefficients at integer exponents are uniformly readable.

10.1 The classical algebraic input

We use the characteristic-zero Newton–Puisseux theorem in the following standard form: if k is algebraically closed of characteristic zero, then $\mathcal{P}(k) = \bigcup_d k((t^{1/d}))$ is algebraically closed. The introductions of Rond and Kedlaya give this formulation and its relation to finite ramification [14, 15]. This is the algebraic-existence input in this second route; the first route in Section 9 uses henselian valuation theory instead.

It follows that $\mathcal{P}(k)$ is real closed when k is real closed. Indeed, $k(i)$ is algebraically closed, and

$$\mathcal{P}(k)(i) = \mathcal{P}(k(i))$$

by separating real and imaginary coefficients at a common denominator. The field $\mathcal{P}(k)$ is ordered by its leading coefficient, hence formally real, and adjoining i makes it algebraically closed. Apply this with $k = \mathbb{R}_c$, using Lemma 8.2.

This classical statement concerns arbitrary coefficient sequences. The following lemma is the additional effective step.

10.2 A coefficientwise effective simple-root lemma

For a polynomial in Y , write $f^{[j]} = f^{(j)}/j!$ for its divided formal derivative. Thus

$$f(Y + H) = \sum_{j \geq 0} f^{[j]}(Y)H^j,$$

where the sum has only finitely many terms.

Lemma 10.1 (Finite-jet lifting). *Let $f(Y) \in \mathbb{R}_c[[u]][Y]$ have uniformly computable coefficients as power series in u . Suppose $y \in \mathbb{R}_c[[u]]$ is a root, with $f'(y) \neq 0$. Then the coefficients of y are uniformly computable. More explicitly, they can be computed uniformly from f and the finite seed data*

$$s = v_u(f'(y)), \quad M > s, \quad a = \sum_{j=0}^{M-1} y_j u^j,$$

where the finitely many coefficients of a are supplied by fast real names.

Proof. Since f and y are integral power series, s is a nonnegative integer. Write $y = a + u^M z$ with $z \in \mathbb{R}_c[[u]]$. For integral arguments, polynomial evaluation is valuation nondecreasing with respect to their difference. In particular,

$$f'(a) - f'(y) \in u^M \mathbb{R}_c[[u]].$$

As $M > s$, this implies $v_u(f'(a)) = s$. Taylor expansion about the actual root gives

$$f(a) = f'(y)(a - y) + \sum_{j \geq 2} f^{[j]}(y)(a - y)^j.$$

The first term has valuation at least $M + s$; every other term has valuation at least $2M > M + s$. Therefore $v_u(f(a)) \geq M + s$.

Consider the transformed polynomial

$$Q(Z) = u^{-(M+s)} f(a + u^M Z). \quad (10.1)$$

Its constant coefficient is integral by the preceding bound. Its linear coefficient is $u^{-s} f'(a)$, an integral unit. For $j \geq 2$, the coefficient of Z^j is

$$u^{(j-1)M-s} f^{[j]}(a),$$

which is divisible by u because $M > s$. Thus $Q \in \mathbb{R}_c[[u]][Z]$, and its reduction modulo u is

$$Q(Z) \bmod u = b + cZ, \quad c \in \mathbb{R}_c \setminus \{0\}. \quad (10.2)$$

All coefficients of Q are uniformly computable from f and the finite seed data: the transformation consists of finite polynomial operations, coefficient shifts, and finite real arithmetic.

There is a unique formal root $z = \sum_{n \geq 0} z_n u^n$ of Q in $\mathbb{R}_c[[u]]$. Its constant term must be $z_0 = -b/c$. Once $z_0, \dots, z_{n-1}$ are known, the coefficient of u^n in $Q(z)$ has the form

$$cz_n + E_n(z_0, \dots, z_{n-1}), \quad (10.3)$$

where E_n is a computably obtainable finite polynomial expression in those coefficients and finitely many coefficients of Q . No nonlinear term can introduce an additional occurrence of z_n at degree n , since all the nonlinear coefficients of Q are divisible by u . Higher terms of its linear coefficient also use only earlier z_j .

Set $z_n = -c^{-1} E_n$ successively. Reciprocation of the nonzero computable real c and every finite arithmetic operation are computable. Recursion on n therefore supplies one algorithm for (z_n) , and uniqueness follows at the same time from (10.3). The previously given $(y - a)/u^M$ is a root, so it is this computed series. Appending the finite prefix a recovers a uniform coefficient algorithm for y . $\square$

Remark 10.2 (Why the finite seed is legitimate). The hypothesis $y \in \mathbb{R}_c[[u]]$ says only that each coefficient individually is a computable real. It does not supply a uniform algorithm for the whole sequence. However, finitely many such coefficients can be equipped with finitely many programs, all hard-coded into one algorithm. Lemma 10.1 uses exactly this finite information to compute every remaining coefficient. It does not hard-code infinitely many programs and does not assume the conclusion it is proving.

Corollary 10.3 (Laurent version). *If $f \in \mathcal{P}_c[Y]$ and $y \in \mathcal{P}(\mathbb{R}_c)$ is a simple root of f , then $y \in \mathcal{P}_c$.*

Proof. Choose a common finite ramification denominator for the finitely many coefficients of f and for y , and write its coordinate as u . Choose an integer $h \geq 0$ making $z = u^h y$ integral. Replacing Y by $u^{-h} Z$ in f and multiplying the polynomial by a sufficiently large power of u gives $g(Z) \in \mathbb{R}_c[[u]][Z]$ with uniformly computable coefficients. The root z remains simple. Apply Lemma 10.1, then shift back by u^{-h} . All choices made here are finite data for the fixed polynomial and root. $\square$

10.3 Algebraic closure inside the full Puiseux field

Theorem 10.4 (Real-closedness of the operational model). *The field $\mathcal{P}_c$ is relatively algebraically closed in $\mathcal{P}(\mathbb{R}_c)$. Consequently $\mathcal{P}_c$ is a real-closed ordered subfield of $\mathbf{No}$. Moreover, $\mathcal{P}_c[i]$ is algebraically closed.*

Proof. Let $y \in \mathcal{P}(\mathbb{R}_c)$ be algebraic over $\mathcal{P}_c$. Choose its minimal polynomial $f \in \mathcal{P}_c[Y]$. This makes sense because Proposition 9.2 has already established that $\mathcal{P}_c$ is a field. Characteristic zero makes the minimal polynomial separable, so y is a simple root. The finitely many coefficients of f admit a uniformly combined finite family of Puiseux coefficient names. Corollary 10.3 yields $y \in \mathcal{P}_c$. This proves relative algebraic closedness.

The ambient field $\mathcal{P}(\mathbb{R}_c)$ is real closed by the classical input above. A positive $x \in \mathcal{P}_c$ has its positive square root in that ambient field; an odd-degree polynomial over $\mathcal{P}_c$ has a real root there. Each such root is algebraic over $\mathcal{P}_c$ and hence belongs to $\mathcal{P}_c$ by the first part. This is the real-closedness criterion. Finally, the standard characterization of real-closed fields says that adjoining a square root of -1 yields an algebraically closed field. $\square$

Corollary 10.5. *Every real root in $\mathbf{No}$ of a nonzero polynomial with coefficients in $\mathcal{P}_c$ is in $\mathcal{P}_c$. Every root in $\mathbf{No}[i]$ of a nonzero polynomial with coefficients in $\mathcal{P}_c[i]$ is in $\mathcal{P}_c[i]$.*

Proof. An ordered extension of a real-closed field has no proper algebraic ordered subextension, and an algebraically closed field has no proper algebraic extension. Alternatively, factor the polynomial over $\mathcal{P}_c[i]$: its full finite list of linear factors leaves no additional roots in a larger field. $\square$

The theorem supplies much more than closure under radicals. For example, all five roots in $\mathbf{No}[i]$ of

$$Y^5 - Y - (1 + t) = 0$$

are coefficient-computable Puiseux surcomplex numbers. It does *not* provide a uniform labeling of these roots from arbitrary coefficient names. Finding the appropriate finite ramification, identifying a root, and certifying a separating finite jet are additional algorithmic tasks.

10.4 Certificates rather than hidden zero tests

The proof suggests a precise interface for an implicit root. A *simple-branch certificate* supplies a ramified coordinate u , an integral polynomial f , an integer s , an integer $M > s$, and a polynomial a with computable real coefficients, with the obligations

$$v_u(f(a)) \geq M + s, \quad v_u(f'(a)) = s. \quad (10.4)$$

These conditions imply that the transform (10.1) reduces to an affine polynomial with nonzero slope. The triangular algorithm then constructs the unique root in $a + u^M \mathbb{R}_c[[u]]$. Thus a certificate can be stated without first supplying an unknown full root.

The original proof shows that every simple formal root has some certificate of this form. It does not show that all certificates can be found or verified uniformly over arbitrary computable real coefficient programs. A proof assistant may validate specific algebraic obligations, and a restricted exact coefficient domain may decide them; neither fact makes the general promise decidable.

Proposition 10.6 (Unit-derivative lifting). *Let $F(u, Y) \in \mathbb{R}_c[[u]][Y]$ have uniformly computable coefficients, and let $c \in \mathbb{R}_c$ satisfy*

$$F(0, c) = 0, \quad F_Y(0, c) \neq 0.$$

Then there is a unique root $y \in \mathbb{R}_c[[u]]$ with constant term c , and its coefficients are uniformly computable from these names and promises.

Proof. Set $y = c + \sum_{n \geq 1} y_n u^n$. At degree $n \geq 1$, the equation $F(u, y) = 0$ is linear in the new unknown y_n , with coefficient $F_Y(0, c)$. All other terms depend on finitely many previously computed coefficients. Division by the fixed nonzero computable real $F_Y(0, c)$ therefore computes y_n . Induction gives existence, uniqueness, and uniform computability. This is the triangular mechanism of Lemma 10.1 without the preparatory change of scale. $\square$

Example 10.7 (Catalan branch). The equation

$$y - t - y^2 = 0, \quad [t^0]y = 0,$$

has unit derivative at $(t, y) = (0, 0)$. Its coefficient recurrence is

$$y_1 = 1, \quad y_n = \sum_{j=1}^{n-1} y_j y_{n-j} \quad (n \geq 2),$$

so

$$y = t + t^2 + 2t^3 + 5t^4 + 14t^5 + 42t^6 + \dots = \frac{1 - \sqrt{1 - 4t}}{2}.$$

The specified constant term chooses the branch. The other root has constant term one and is not an alternative output for this branch certificate.

Example 10.8 (A derivative that is not a unit). Take

$$f(Y) = Y^2 - u^2(1 + u), \quad a = u, \quad s = 1, \quad M = 2.$$

Then $v_u(f(a)) = 3 = M + s$ and $v_u(f'(a)) = 1 = s$. The certified transform is

$$u^{-3}f(u + u^2Z) = -1 + 2Z + uZ^2.$$

Its reduction has the unique root $Z = 1/2$. Lifting yields

$$y = u + \frac{1}{2}u^2 - \frac{1}{8}u^3 + \frac{1}{16}u^4 - \frac{5}{128}u^5 + \dots = u\sqrt{1 + u}.$$

Thus the finite-jet method applies even when the original derivative has positive valuation. The required condition is a sufficiently accurate seed relative to that valuation, not an original unit derivative.

Coefficient-field generality. The finite-jet transformation is algebraic over any characteristic-zero coefficient field. Its recursion uses uniform ring operations and the inverse of only one nonzero slope. Thus if the coefficient field itself has only extensional inverse closure, fix one name for that inverse as additional finite seed data. This observation, emphasized in B, is exactly what the iterated Puiseux proof needs; infinitely many independent inverse choices would not establish a uniform coefficient sequence.

11 Effective completion and the role of a modulus

An ordinary valuation-Cauchy sequence eventually stabilizes every bounded initial coefficient section. To turn this into an algorithm, one must know *when* stabilization has occurred.

Definition 11.1 (Effective valuation-Cauchy sequence). A uniformly numerically named sequence (x_n) has a computable valuation modulus if there is a total computable function $\mu : \mathbb{Q} \rightarrow \mathbb{N}$ such that

$$n, m \geq \mu(B) \implies v(x_n - x_m) \geq B. \quad (11.1)$$

This is an exact stabilization assertion for coefficients below B , not a bound on their ordinary real approximation errors.

Theorem 11.2 (Effective sequential completeness). *From a uniformly named sequence in $\mathcal{L}_c$ and a computable valuation modulus, one can compute its unique left-finite valuation limit as an element of $\mathcal{L}_c$.*

Proof. At exponent q , define the limiting coefficient by

$$a_q = [t^q]x_{\mu(q+1)}. \quad (11.2)$$

All later terms agree there, by the modulus at $q + 1$. Thus this formula supplies a joint Cauchy coefficient algorithm.

Let $n_0 = \mu(0)$ and let ℓ_0 be the supplied lower bound of x_{n_0} . Set $\ell = \min(\ell_0, 0)$. Every limiting coefficient below zero agrees with x_{n_0} , so it vanishes below ℓ . More explicitly, for any $q < 0$, choose an index later than both $\mu(0)$ and $\mu(q + 1)$; the two stabilization promises identify their coefficients at q .

For a threshold B , take $F_{x_{\mu(B)}}(B)$, filter it to $[\ell, B)$, and use the resulting finite list as the limit cover. Every limiting coefficient below B agrees with $x_{\mu(B)}$: compare both to an index later than $\mu(B)$ and $\mu(q + 1)$. Hence the list is complete. The limit has a numerical name, and for $n \geq \mu(B)$ it satisfies $v(x - x_n) \geq B$. Uniqueness follows by comparing every rational coefficient, or by taking arbitrarily large B in the ultrametric inequality. $\square$

Theorem 11.3 (Effective completion characterization). *Every element of $\mathcal{L}_c$ is the limit of a uniformly computable sequence of finite Puiseux polynomials with a computable valuation modulus. Conversely, every such effective valuation limit of uniformly named $\mathcal{P}_c$ elements belongs to $\mathcal{L}_c$.*

Proof. Given $x \in \mathcal{L}_c$, take $p_n = J_n x$. Its computed exponent list is finite, so the least common multiple of its denominators supplies a Puiseux grid. The real coefficients have Cauchy names uniformly inherited from x . Proposition 6.4 gives $v(x - p_n) \geq n$, hence $v(p_n - p_m) \geq \min(n, m)$. Thus $\mu(B) = \max(0, \lceil B \rceil)$ is a modulus. The converse follows by translating Puiseux names to numerical names and applying Theorem 11.2. $\square$

This is the precise completion analogy with computable real numbers in the present theory. The finite approximants retain Cauchy-named real coefficients, so it is a *two-layer* completion, not the completion of a discretely coded rational-monomial field in its valuation metric. Rational constants cannot approximate an arbitrary irrational constant in that metric, because every nonzero real difference has valuation zero.

Proposition 11.4 (A computable Cauchy sequence without a computable limit). *There is a uniformly computable sequence of rational polynomials in t which is valuation-Cauchy but whose left-finite limit has no numerical name. It has no computable valuation modulus.*

Proof. At stage s put

$$z_s = \sum_{\substack{e \leq s \\ \text{machine } e \text{ halts by stage } s}} t^e.$$

Each polynomial is computed by finitely many simulations. For a fixed threshold B , only finitely many integers $e < B$ matter. Each one either never halts or eventually has its coefficient permanently set to one. Hence the sequence is valuation-Cauchy, with left-finite limit

$$z = \sum_{e \geq 0} \chi_K(e) t^e.$$

A numerical name for z would compute its zero-one coefficients and therefore decide K . If a computable modulus existed, Theorem 11.2 would produce just such a name. $\square$

Even a *computable* modulus does not keep a sequence inside $\mathcal{P}_c$: the finite partial sums of (9.3) have an obvious computable modulus and converge to $u \in \mathcal{L}_c \setminus \mathcal{P}_c$. The support cover, rather than a single denominator, is what survives effective completion.

11.1 A different limit theorem: coefficientwise real convergence

Theorem 11.5 (Uniform coefficientwise limits on a fixed lattice). *Suppose $x_j = \sum_{n \geq N} a_{j,n} u^n$, $u = t^{1/d}$, share fixed d, N , the real coefficients are uniformly computable in (j, n) , and a computable function $M(n, k)$ satisfies*

$$|a_{j,n} - a_{\ell,n}| \leq 2^{-k} \quad (j, \ell \geq M(n, k)).$$

Then $a_n = \lim_j a_{j,n}$ is uniformly computable, and $\sum_{n \geq N} a_n u^n$ has a uniformly constructed Puiseux name.

Proof. For (n, k) choose $j = M(n, k + 1)$ and approximate $a_{j,n}$ with error $2^{-(k+1)}$. Its distance from the limit is at most $2^{-(k+1)}$ by real completeness. The combined error is at most 2^{-k} . Fixed metadata certify the support of the result regardless of cancellation. $\square$

This is not exact valuation stabilization: rational constants converging to an irrational real never have positive valuation error. Without a modulus, stagewise rational approximations to r_K from Proposition 3.5 converge coefficientwise to a noncomputable constant. Conversely, the exact-valuation example with increasingly many halting coefficients has eventually *equal* low coefficients but no computable stabilization modulus. The two failures should not be identified. Nor should $\mathcal{L}_c$ be called sequentially complete without qualification: the computable-modulus theorem is strictly weaker than closure under every classically Cauchy sequence of its elements.

12 Sharp limits of uniform computation

The field property and the effective algorithms already proved might suggest that a missing normalization subroutine is merely a technical inconvenience. The next results show an unavoidable obstruction.

12.1 A uniformly computable real family encoding halting

Fix a machine enumeration with halting stages numbered from one. Define

$$a_e = \begin{cases} 2^{-s}, & \text{machine } e \text{ first halts at stage } s, \\ 0, & \text{machine } e \text{ never halts.} \end{cases} \quad (12.1)$$

This is a uniformly Cauchy-computable family of nonnegative reals. To approximate a_e to error 2^{-k} , simulate for k stages. If a halt has occurred, return its value; otherwise return zero. A later halt contributes less than 2^{-k} .

Theorem 12.1 (Valuation, inversion, and sign obstructions). *On raw numerical names, and already on raw Puiseux names:*

- (i) *the valuation of a nonzero element is not uniformly computable;*
- (ii) *reciprocation on nonzero inputs is not uniformly computable;*
- (iii) *strict positivity and strict negativity are not semidecidable in general, even on inputs promised nonzero;*
- (iv) *equality is a Π_1^0 -complete promise problem on valid indices, although nonzeroness and disequality are semidecidable.*

Proof. For (i) use the always positive nonzero family $x_e = a_e + t$. It has a fixed denominator-one name with support cover contained in $\{0, 1\}$, but

$$v(x_e) = \begin{cases} 0, & e \in K, \\ 1, & e \notin K. \end{cases}$$

Thus computing the valuation would decide halting.

For (ii), the coefficient of the inverse at exponent -1 satisfies

$$[t^{-1}](a_e + t)^{-1} = \begin{cases} 0, & a_e > 0, \\ 1, & a_e = 0. \end{cases} \quad (12.2)$$

When $a_e = 0$ the inverse is t^{-1} . When $a_e > 0$ its formal expansion is $a_e^{-1} \sum_{n \geq 0} (-t/a_e)^n$, which has no negative exponent. A hypothetical inverse compiler followed by coefficient approximation to error $1/8$ would decide K .

For (iii), put $y_e = a_e - t$. This is never zero, and $y_e < 0$ exactly when $e \notin K$. If negativity were semidecidable, the complement of K would be computably enumerable. Applying the same argument to $-y_e$ excludes semidecidable positivity.

For (iv), equality to zero on the constants a_e would decide whether $e \notin K$. In contrast, to semidecide $x \neq 0$, enumerate rational exponents and precision levels in parallel. A coefficient is certified nonzero whenever $|A(q, k)| > 2^{-k}$. A nonzero coefficient eventually produces such a certificate, while a zero series never does. Applying this to a difference semidecides disequality ($x \neq y$). Equality is the absence of such a witness; the constants a_e give Π_1^0 -hardness. This classification concerns promised valid indices, not the additional totality and consistency conditions for arbitrary programs. $\square$

An important feature is that the inverse obstruction uses *only* nonzero, strictly positive inputs. Supplying a Boolean assertion “nonzero” or “positive” does not replace a leading-exponent certificate.

Theorem 12.2 (A representation-independent incompatibility). *Let F be any represented subfield of a Hahn field containing the real constants and a positive infinitesimal monomial t . Suppose that:*

- (a) *ordinary real Cauchy names can be mapped uniformly to constant names;*
- (b) *$a \mapsto a + t$ is computable on those names;*
- (c) *the coefficient at t^{-1} of an output element is uniformly readable as a Cauchy real.*

Then no uniform reciprocal operation on all nonzero names can exist. The conclusion already holds when the input values are the degree-one polynomials $a + t$ with computable nonnegative real a .

Proof. Apply (a) and (b) to the uniform family (12.1). An inverse operation and (c), followed by (12.2), would decide halting. $\square$

Enlarging the permitted support order types therefore cannot remove this particular obstruction. A successful alternative representation must change one of the capabilities: supply normalization data, weaken coefficient access, use a more informative real representation, or treat some symbolic operations as denotational rather than coefficient-computable.

12.2 Exactly what one halting oracle buys

Theorem 12.3 (Normalization has halting degree on computable indices). *Given program indices for a valid computable numerical name, with nonzero value promised, the halting oracle $0'$ suffices to compute its leading exponent and to select an ordinary computable index for its inverse. Conversely, a solver for either of these uniform tasks decides the halting problem on computable inputs.*

Proof. First search, without an oracle, for a certified nonzero coefficient. The semidecision procedure in Theorem 12.1 terminates because $x \neq 0$. Let q be its exponent. The finite cover $F(q+1)$ contains all nonzero coefficients at or below q and only finitely many other candidates. For each candidate r , the predicate $a_r \neq 0$ is computably enumerable uniformly from the input program index. A halting oracle decides all these finitely many predicates. Taking the least nonzero candidate yields the true leading exponent λ .

Now hard-code this rational λ into the inverse compiler of Theorem 7.4. The resulting inverse index needs no oracle when it is subsequently run. The lower bounds are the explicit reductions using $a_e + t$ in Theorem 12.1. $\square$

The qualification “on computable indices” is essential. A fixed oracle $0'$ does not decide the zero status of arbitrary real coefficients supplied by arbitrary noncomputable oracle names. In the relativized formulation, the corresponding jump of the input oracle is used. We are not assigning a more delicate Weihrauch degree to these operators here.

12.3 Square roots, arbitrary root selection, and modulus

Theorem 12.4 (Root-selection obstructions). *Neither positive square root on all positive bare names nor selection of any root of every separable quadratic is uniformly computable in $\mathcal{P}_c$ or $\mathcal{L}_c$. The first assertion holds even when a common positive lower bound t is given.*

Proof. For $x_e = a_e + t \geq t$, if $a_e = 0$ its positive root is $t^{1/2}$. If $a_e > 0$ its positive root is $\sqrt{a_e} \sum_{n \geq 0} \binom{1/2}{n} (t/a_e)^n$, with integer exponents only. The coefficient at $t^{1/2}$ is therefore one or zero according as e fails to halt or halts. Reading this coefficient decides halting.

For the stronger branch-independent assertion from B, consider

$$Q_e(Y) = Y^2 - (a_e^2 + t^2).$$

This polynomial is separable even when $a_e = 0$. In that case its two roots are $\pm t$, with coefficient at t equal to ± 1 . If $a_e > 0$, both roots are $\pm a_e(1 + t^2/a_e^2)^{1/2}$ and their coefficient at t is zero. An approximation with error less than $1/4$ to the coefficient of *any* selected root distinguishes the cases. The same argument applies when the selector is allowed surcomplex outputs, since these quadratics already split with exactly those two real roots. $\square$

Proposition 12.5 (Certified square root and the extended halting threshold). *For a positive numerical name with known true leading exponent λ , positive square root is uniformly computable. On ordinary computable indices, $0'$ is necessary and sufficient for equality, sign, valuation on nonzero values, inverse on nonzero values, and positive square root on positive values. After selecting finite leading data, the inverse and square-root outputs can be ordinary computable programs.*

Proof. Let $a = [t^\lambda]x > 0$ and $h = a^{-1}t^{-\lambda}x - 1$. Then h has positive support and a computable gap $\delta > 0$, obtained from its finite cover below one. Set

$$\sqrt{x} = \sqrt{a} t^{\lambda/2} \sum_{n \geq 0} \binom{1/2}{n} h^n.$$

Only finitely many powers can contribute below a given cutoff, since their valuations are at least $n\delta$. Finite convolution and cover unions give a valid numerical name. On a Puiseux grid this just doubles the denominator if necessary and otherwise preserves the coefficient lattice. The formal binomial identity proves that the result is the positive root.

A halting oracle decides the nonzeroness search for a whole series and for any individual coefficient. Theorem 12.3 then finds the leading exponent. Its nonzero real coefficient has a computable strict sign, and the preceding construction gives the square root. The negative families already displayed prove every lower bound. No upper bound for arbitrary polynomial branch selection is asserted here. $\square$

Leading data for two inputs do not automatically normalize their sum: leading coefficients may cancel. Thus ring constructors return redundant names; normalization is a separate task. A leading certificate is a mathematical promise or a checked proof in a stated proof system, not a universally decidable data-validation procedure.

12.4 Exact coefficient domains change the uniformity boundary

Suppose instead that coefficients lie in a computable ordered field k with a decidable exact zero test, such as $\mathbb{Q}$ or the real algebraic numbers with their usual exact descriptions. Keep the same effective finite-section covers, but return exact coefficient codes.

Proposition 12.6 (Exact coefficients permit leading-term search). *For this exact-coefficient representation, the leading exponent and sign of a nonzero series are computable uniformly. Reciprocation is uniformly computable on nonzero names. Equality of arbitrary infinite series still need not be decidable.*

Proof. Inspect the covers $F(0), F(1), F(2), \dots$ and their coefficients until a nonzero coefficient is found. Every rational exponent is eventually below one of these integer thresholds. At that stage, sort the finite cover and choose its first nonzero entry; this is the true leading exponent because all earlier possible nonzero positions are included. Exact order on k gives its sign, and the geometric inverse construction is now uniform.

For equality, from a machine e build the series whose coefficient at integer $s \geq 1$ is one exactly when e first halts at stage s , and whose other coefficients are zero. Coefficient access is a finite computation, and the integer grid supplies covers. The series is zero exactly when the machine never halts. Thus a uniform equality decider would decide halting. $\square$

With k equal to the real algebraic numbers the same Hensel and valuation arguments prove an extensionally real-closed field. What is lost is the uniform inclusion of arbitrary ordinary computable reals as constants, not just algebraic constants. With $k = \mathbb{Q}$ the field is not real closed, since a square root of the constant two cannot have rational coefficients. These alternatives clarify which capabilities a symbolic system is buying with an exact coefficient type.

12.5 Countable does not mean effectively enumerable

Proposition 12.7 (A coefficient diagonal). *No uniformly computable sequence of valid numerical names enumerates all of $\mathcal{L}_c$, even up to equality. The same is true for $\mathcal{P}_c$.*

Proof. Let (x_n) be such a sequence. Compute a rational r_n within $1/8$ of $[t^n]x_n$, and choose

$$b_n = \begin{cases} 1, & r_n \leq 1/2, \\ 0, & r_n > 1/2. \end{cases}$$

In the first case the true coefficient is at most $5/8$, and in the second it is greater than $3/8$. Thus in either case it differs from b_n by at least $3/8$. The series $y = \sum_{n \geq 0} b_n t^n$ has a computable Puiseux name, but $y \neq x_n$ for every n by its coefficient at exponent n . This proves both claims. $\square$

The diagonal uses an approximation with a safety margin, not an equality test on reals. It also shows why listing all programs and silently keeping only the valid ones is not an executable enumeration strategy.

13 Infinitesimal functional calculus and formal integration

13.1 Formal series need support bounds, not real convergence radii

Theorem 13.1 (Effective infinitesimal evaluation). *Let $f(Z) = \sum_{n \geq 0} c_n Z^n$ have uniformly Cauchy-computable real coefficients. If $h \in \mathcal{L}_c$ is numerically named with positive support, then the Hahn evaluation*

$$f(h) = \sum_{n \geq 0} c_n h^n$$

has a uniformly computable numerical name. If $h \in \mathcal{P}_c$, the result belongs to $\mathcal{P}_c$ and its name can be computed on the same exponent grid.

Proof. Extract $\delta > 0$ with $v(h) \geq \delta$. A nonzero real c_n has valuation zero, so $v(c_n h^n) \geq n\delta$. Below any threshold B , only terms with $n\delta < B$ can contribute. Compute their finite covers by the ring operations and take the union, after discarding negative candidates as justified by the nonnegative support. At a queried exponent q , choose N with $(N+1)\delta > q$ and compute its coefficient in the finite sum through N . This gives a total joint coefficient algorithm and a lower bound zero. For a Puiseux input, finite sums, powers, and the stabilized coefficients remain on its fixed grid. $\square$

In particular $\exp(h)$, $\log(1+h)$, $\sin(h)$, $\cos(h)$, and $(1+h)^a$ are computable whenever their formal coefficients are given by the usual computable real formulas. The exponent a in the binomial series may itself be a Cauchy-computable real:

$$(1+h)^a = \sum_{n \geq 0} \binom{a}{n} h^n, \quad \binom{a}{n} = \frac{a(a-1) \cdots (a-n+1)}{n!}.$$

For each coefficient, identities between these formal series reduce to finite formal coefficient identities. They do not require convergence in the full surreal order topology.

For a finite $x = r + h \in \mathcal{L}_c$ with h infinitesimal and $r = \text{st}(x)$, one can define and compute

$$\exp_{\text{fin}}(x) = e^r \exp(h).$$

If x has positive real residue r , then $\log_{\text{unit}}(x) = \log r + \log(1 + (x-r)/r)$ is likewise computable under that promise. These formulas are local, coefficientwise extensions of ordinary real functions. No uniform computation of the unrestricted surreal exponential, logarithm, or trigonometry at arbitrary infinite arguments is asserted. Broader exponential and differential surreal subfields involve additional closure questions and representations; see, for example, [11].

Example 13.2 (A ramified square root with exact infinitesimal terms). Hensel lifting, or the formal binomial identity, gives

$$\sqrt{\omega^2 + \omega} = \omega + \frac{1}{2} - \frac{1}{8\omega} + \frac{1}{16\omega^2} - \frac{5}{128\omega^3} + \cdots.$$

The positive branch is fixed by writing the result as $\omega(1+t)^{1/2}$. For any exponent threshold, only finitely many binomial coefficients are needed, and the remainder has a certified valuation bound. The omitted terms are not an unspecified real error.

13.2 A computable derivation and an exact obstruction to primitives

Define a formal t -derivation by

$$D_t \left(\sum_q a_q t^q \right) = \sum_q q a_q t^{q-1}. \quad (13.1)$$

The coefficient field is held constant. This convention is important: $D_t(\omega) = -\omega^2$, whereas the usual normalized Berarducci–Mantova derivation has $D(\omega) = 1$. We are not silently identifying these two derivations or claiming a new construction of the latter.

Theorem 13.3 (Effective differentiation and the formal residue obstruction). *The map $D_t : \mathcal{L}_c \rightarrow \mathcal{L}_c$ is a uniformly computable derivation, with kernel $\mathbb{R}_c$. Define*

$$\text{res}_t(x) = [t^{-1}]x.$$

There is a uniformly computable linear map $I_0 : \mathcal{L}_c \rightarrow \mathcal{L}_c$ satisfying

$$D_t I_0(x) = x - \text{res}_t(x) t^{-1}, \quad [t^0] I_0(x) = 0. \quad (13.2)$$

Consequently x has a primitive in $\mathcal{L}_c$ exactly when $\text{res}_t(x) = 0$; under that promise $I_0(x)$ is the unique primitive with zero constant coefficient. The same statements hold in $\mathcal{P}_c$.

Proof. The coefficient at r of $D_t x$ is $(r+1)a_{r+1}$. Shift a cover for x below $B+1$ down by one and use lower bound $\ell_x - 1$; optional deletion of the exponent arising from $q=0$ is justified exactly. This is a numerical name. For each coefficient of a product, the convolution sum is finite. The identity $r+s$ for the derivative multiplier at a product term splits as the sum of the two derivative multipliers, proving the Leibniz rule. Characteristic zero implies that every coefficient at a nonzero input exponent must vanish in the kernel. Thus the kernel consists exactly of real constants in $\mathcal{L}_c$, namely $\mathbb{R}_c$.

Set

$$I_0(x) = \sum_{q \neq -1} \frac{a_q}{q+1} t^{q+1}. \quad (13.3)$$

The coefficient at exponent $r \neq 0$ is a_{r-1}/r , and the coefficient at zero is set to zero. Shifting $F_x(B-1)$ by one and deleting zero provides a finite cover, with lower bound $\ell_x + 1$. Thus the construction is uniform. Direct coefficient calculation gives (13.2).

For every y , the coefficient at -1 of $D_t y$ is zero, since it could only come from the exponent-zero term, multiplied by zero. This proves necessity of the residue condition; (13.2) proves sufficiency and the kernel statement proves uniqueness. Rational shifting preserves a fixed Puiseux grid, proving the final assertion. $\square$

The condition $\text{res}_t(x) = 0$ is not generally decidable from a raw numerical name, but the operator I_0 itself is total and computable. Thus a useful interface can always return a primitive-with-obstruction pair $(I_0(x), \text{res}_t(x))$ without deciding whether the obstruction vanishes. For $x = t^{-1}$ the obstruction is one; a primitive would require a logarithmic extension rather than another left-finite rational-power series.

13.3 The full residue obstruction and a differential limitation

Corollary 13.4 (Kernel, image, and quotient of the formal derivative). *For $F = \mathcal{P}_c$ or $F = \mathcal{L}_c$, let $D = D_t$ and let I_0 be the zero-constant primitive operator omitting the exponent -1 . Then*

$$I_0 D x = x - [t^0]x, \quad D I_0 x = x - \text{res}_t(x)t^{-1}, \quad \ker D = \mathbb{R}_c,$$

where $\text{res}_t(x) = [t^{-1}]x$. Moreover,

$$D(F) = \ker(\text{res}_t), \quad F/D(F) \cong \mathbb{R}_c$$

as $\mathbb{R}_c$ -vector spaces. Every primitive of an integrable x is $I_0 x + c$, $c \in \mathbb{R}_c$. Deciding whether a primitive exists is not computable on raw names.

Proof. The coefficient identities follow from $D(t^q) = qt^{q-1}$ and division by $q+1$ away from $q = -1$. They prove both the kernel and image descriptions; residue is onto since ct^{-1} has residue c . Two primitives differ by a member of the kernel. Finally ct^{-1} is integrable exactly when the computable real c is zero, giving the decision obstruction. $\square$

Here $D(F)$ denotes the additive image, not an ideal of fields; the quotient is a vector-space quotient. A logarithm symbol could supply a primitive of t^{-1} in a larger differential field, but does not make that primitive an element of the present coefficient field.

Proposition 13.5 (A simple differential equation outside the workspace). *The equation $Dy = -t^{-2}y$ has no nonzero solution in $\mathcal{P}_c$ or $\mathcal{L}_c$.*

Proof. For $y \neq 0$, termwise differentiation gives $v(Dy) \geq v(y) - 1$, allowing infinity when $Dy = 0$. The proposed right side has valuation $v(y) - 2$, a contradiction. $\square$

Thus real closedness and effective infinitesimal calculus do not imply closure under arbitrary differential equations. In a suitable larger exponential differential field, $\exp(t^{-1})$ is the expected kind of solution; it is not licensed here by the positive-support substitution rule.

14 Surcomplex computability

Define the numerically computable surcomplex field by

$$\mathcal{L}_c^{\mathbb{C}} = \mathcal{L}_c[i] = \{x + iy : x, y \in \mathcal{L}_c\}.$$

A name is a pair of numerical names. Equivalently, one can use uniformly Cauchy-computable complex coefficients and the union of the two real support covers. The complex valuation is $v(x + iy) = \min(v(x), v(y))$, with the usual zero convention.

Proposition 14.1 (Effective complexification). *The two naming conventions are uniformly interconvertible. Addition, multiplication, conjugation, real and imaginary parts, and coefficient extraction are computable. The field is algebraically closed and has residue field $\mathbb{C}_c = \mathbb{R}_c[i]$. Reciprocation exists extensionally but remains nonuniform on unrestricted nonzero names.*

Proof. Convert between paired real oracles and complex coefficient oracles componentwise. Support covers are united in one direction and reused in the other, since a complex coefficient is zero exactly when both its real and imaginary parts are zero. The formulas

$$(x + iy)(u + iv) = (xu - yv) + i(xv + yu), \quad \overline{x + iy} = x - iy$$

use only the uniform real ring operations. Algebraic closedness is Theorem 8.4. Coefficient zero gives the complex residue homomorphism with residue field $\mathbb{C}_c$. For $z \neq 0$, $z^{-1} = \bar{z}/(x^2 + y^2)$ belongs to the field by extensional real field closure. A uniform inverse algorithm would restrict to the real input family $a_e + t$ and contradict Theorem 12.2. $\square$

A supplied true leading exponent and its nonzero complex coefficient restore the geometric inverse algorithm: ordinary complex reciprocation is computable with the nonzero promise. Complex simple residue roots lift by precisely the Newton proof in Theorem 8.1, since the complex leading coefficient of each derivative remains the same nonzero residue. Formal evaluation and D_t, I_0, res_t extend componentwise.

This gives a substantial exact surcomplex algebra, including algebraic solutions, infinitesimal holomorphic power series, and residue-controlled formal primitives. It is not a theorem that arbitrary contours or analytic functions on the whole surcomplex plane have effective representations. Such claims would require specifying function names, support dependencies, and contour conventions separately.

Proposition 14.2 (Squared modulus is uniform; modulus is not). *On pair names in $\mathcal{P}_c[i]$ and $\mathcal{L}_c[i]$, $|z|^2$ is uniformly computable. Every $|z|$ belongs to the real base field, but the map $z \mapsto |z|$ is not uniformly computable, even on nonzero real inputs.*

Proof. For $z = x + iy$, $|z|^2 = x^2 + y^2$ uses ring operations. Real closedness supplies its nonnegative square root. For $z_e = t - a_e$, the coefficient at t in $|z_e|$ is 1 when $a_e = 0$, and -1 when $a_e > 0$, since then $|z_e| = a_e - t$. Uniformly reading that output coefficient would decide K . $\square$

Certified nonzero leading data permit inverse and modulus constructions, but adjoining i does not restore a missing normalization algorithm. Formal complex substitution is componentwise and support-controlled; no contour, global analytic continuation, or unrestricted global exponential theorem follows from algebraic closedness alone.

15 Beyond one scale: an iterated numerical atlas

The rational-exponent core is useful but omits ω^ω and many other structurally simple numbers. It can be extended without abandoning ordinary computable real constants. The following construction makes that claim precise rather than treating a higher-rank Hahn field as automatically effective.

15.1 An effective left-finite extension of a represented field

Suppose k is a countable ordered field all of whose elements have computable names, with uniform named ring operations and extensional inverse closure. Define $\text{ELF}(k; u)$ by series

$$X = \sum_{q \in \mathbb{Q}} a_q u^q,$$

where a program, given q , returns a k -name for a_q , together with a rational lower bound and computable finite covers of the nonzero outer support. At the bottom level, returning a real name means returning an index for a Cauchy program. At higher levels it means returning an index and finite data for a name at the preceding level. Correctness and totality are again semantic promises.

Finite sums and products use exactly the cover formulas from Section 7, with finite coefficient arithmetic in k . For inversion of a particular nonzero X , hard-code its leading rational exponent and a computable name for the inverse of its leading coefficient in k . The geometric construction then works uniformly with those data. This extra coefficient-inverse datum cannot be omitted when k itself has nonuniform inversion.

Theorem 15.1 (Iterated numerically conservative real-closed fields). *Put $\mathcal{L}_0 = \mathbb{R}_c$ and inductively*

$$\mathcal{L}_d = \text{ELF}(\mathcal{L}_{d-1}; t_d), \quad t_d = \omega^{-\omega^{d-1}} \quad (d \geq 1),$$

with t_d treated as an outer infinitesimal relative to $\mathcal{L}_{d-1}$. Then each $\mathcal{L}_d$ embeds as a countable real-closed subfield of $\mathbf{No}$; $\mathcal{L}_1 = \mathbb{R}_c$; and the constant embeddings give

$$\mathcal{L}_0 \subset \mathcal{L}_1 \subset \mathcal{L}_2 \subset \cdots.$$

Their union $\mathcal{L}_{<\omega}$ is a countable real-closed subfield with $\mathcal{L}_{<\omega} \cap \mathbb{R} = \mathbb{R}_c$. It contains every $\omega^{\omega^{d-1}}$ for positive integer d .

Proof. Proceed by induction on d . The coefficient expansion of an element of $\mathcal{L}_d$ flattens to a real Hahn series whose exponents lie in

$$\Gamma_d = \mathbb{Q} + \mathbb{Q}\omega + \cdots + \mathbb{Q}\omega^{d-1} \subseteq \mathbf{No}.$$

The order on Γ_d is lexicographic with its highest-degree coordinate most significant. Each outer support is well ordered, and for each of its exponents the inner coefficient support is well ordered. A nonempty subset of the flattened support has a least outer coordinate, then a least inner exponent. Induction gives well-ordering of the flattened support. Uniqueness of the coordinate decomposition in Γ_d and Hahn arithmetic show that flattening is an ordered-field embedding into $\mathbb{R}((\omega^{-\Gamma_d})) \subseteq \mathbf{No}$.

For clarity, inner exponents need not be bounded by real numbers. What matters is that every inner exponent is smaller in magnitude than a positive rational multiple of ω^{d-1} . Thus t_d is indeed smaller than every positive element of the previously constructed field, as the outer-series interpretation requires.

The finite-cover algorithms just described give uniform ring operations and extensional field closure. Countability follows by induction from countably many program indices. For the outer valuation, the value group is $\mathbb{Q}$ and the residue field is $\mathcal{L}_{d-1}$, and its valuation ring is convex for the lexicographic order.

We verify henselianity with special attention to the possible nonuniformity in the coefficient field. Let P have integral outer coefficients and let $c \in \mathcal{L}_{d-1}$ be a simple residue root. Its derivative residue $a \in \mathcal{L}_{d-1}$ is nonzero. By the inductive field property, select once and for all a computable $\mathcal{L}_{d-1}$ -name for a^{-1} . Every Newton derivative then has the same outer leading coefficient a . Hence every outer unit inverse can be computed with the same selected name for a^{-1} , and no fresh nonuniform coefficient choices are needed at later iterations. The positive-gap and doubling proof of Theorem 8.1 produces uniformly named stabilized outer coefficients and a finite-cover algorithm for the root. This proves henselianity. The inductive real closedness of $\mathcal{L}_{d-1}$ and Lemma 8.3 give real closedness of $\mathcal{L}_d$.

The constant embedding at each stage is effective. Every finite set of coefficients from the union lies in some $\mathcal{L}_d$, where the requisite positive square roots and odd-degree roots exist; hence the union is real closed. Uniqueness of flattened Hahn normal forms shows that a real constant in $\mathcal{L}_d$ must already have been a real constant in $\mathcal{L}_{d-1}$. Induction gives the real-slice assertion. Finally t_d^{-1} supplies $\omega^{\omega^{d-1}}$. $\square$

This atlas goes beyond a single fixed rank while retaining numerical conservativity. It still does not cover all surreal scales or all structurally computable surreals. Its exponent groups are finite rational polynomials in ω ; for example, ω^{ω^ω} is not in this particular union, by normal-form uniqueness. Adding a new scale is a mathematical enlargement of the workspace, not an implicit coercion that changes the existing exponent order.

The same construction works for complexifications. Each $\mathcal{L}_d[i]$ and their union are algebraically closed. No uniform reciprocal claim is restored by passing to this union: ordinary real inputs and the coefficient of the original t_1^{-1} still give the obstruction of Theorem 12.2.

15.2 The parallel Puiseux hierarchy and the common oracle bound

Define $\mathcal{P}_0 = \mathbb{R}_c$ and, with the same scales t_d , let $\mathcal{P}_d$ consist of lower-bounded outer Puiseux series whose coefficient names in $\mathcal{P}_{d-1}$ are generated by one total ordinary program. The outer denominator is fixed for each name, not for the entire level. Thus $\mathcal{P}_1 = \mathcal{P}_c$, whereas $\mathcal{L}_1 = \mathcal{L}_c$.

Theorem 15.2 (Two compatible finite-rank hierarchies). *For every $d \geq 1$, $\mathcal{P}_d$ is a countable real-closed subfield of $\mathbf{No}$, with real trace $\mathbb{R}_c$, and there is a uniform embedding of names $\mathcal{P}_d \hookrightarrow \mathcal{L}_d$. This inclusion is proper. Both hierarchies have natural value group $\Gamma_d = \mathbb{Q} + \mathbb{Q}\omega + \cdots + \mathbb{Q}\omega^{d-1}$, or equivalently lexicographic $\mathbb{Q}^d$. Their outer valuation instead has value group $\mathbb{Q}$ and residue the preceding level. Their finite-level unions are real closed and preserve exactly the real trace $\mathbb{R}_c$.*

Proof. The field and embedding arguments use common outer denominators, finite convolution, and the reciprocal recurrence, choosing once the inverse of the leading lower-level coefficient. Every outer lattice supplies a finite cover, and induction converts its inner names. To prove real closedness, apply classical Newton–Puiseux over the preceding real-closed field and the finite-jet argument of Section 10. A fixed finite jet and one named inverse slope give a uniform tail over that coefficient field. Separability and the minimal-polynomial argument finish the induction. Alternatively, the outer simple-Hensel argument stays on a common grid.

Flattening and zero-coordinate extraction prove the valuation and trace statements exactly as in Theorem 15.1. Proper inclusion is witnessed at level d by $\sum_{n \geq 1} t_d^{n+1/(n+1)}$: it has effective left-finite outer covers but no bounded outer denominator. Inner coefficients cannot absorb a nonzero outer coordinate. Finally every polynomial over either union has its finitely many coefficients in one finite level, where the required algebraic roots already exist. $\square$

Proposition 15.3 (Finite rank does not require iterated jumps). *At every finite level of either hierarchy, nonzeroness is semidecidable. On valid ordinary computable indices, $0'$ is necessary and sufficient for sign, leading exponent vector, inverse, and positive square root on their appropriate nonzero or positive domains. Oracle-assisted inverse and root construction can return ordinary programs.*

Proof. Dovetail searches through rational coefficient addresses down the finite nesting depth until a nonzero real leaf is detected. This halts precisely for nonzero series, so $0'$ decides zero at every

level. Find any nonzero outer coefficient, use a finite cover to find the least such coefficient (or scan the discrete Puiseux grid), then descend to its leading inner coefficient with the same oracle. The inverse and binomial recursions use only finitely selected lower-level inverse or root names and uniform ring operations. The nesting depth decreases, so the recursion terminates; it does not call the jump of $0'$. The rank-one reductions embedded as constants or as expressions in t_1 give the lower bounds. $\square$

The distinction between the outer valuation and the full natural valuation prevents a misleading statement that every higher-rank residue field is $\mathbb{R}_c$: only the full natural residue is $\mathbb{R}_c$. Finite rank enlarges the scale geometry, not the ordinary computational power of a name.

16 Oracle information and support geometry are independent

For an oracle $X \subseteq \mathbb{N}$, write $\mathbb{R}_X$ for the reals with X -computable fast names. Define $\mathcal{P}_X$ and $\mathcal{L}_X$ by relativizing, respectively, the Puiseux names and the coefficient-and-cover names to X . The oracle subscript is an oracle set, not the finite level index d used in the preceding section. In particular $\mathcal{P}_\emptyset = \mathcal{P}_c$ and $\mathcal{L}_\emptyset = \mathcal{L}_c$.

Theorem 16.1 (Relativization and exact degree embeddings). *For every X , both fields are countable and real closed, have real trace $\mathbb{R}_X$, value group $\mathbb{Q}$, and residue field $\mathbb{R}_X$. For arbitrary oracles X, Y ,*

$$\mathcal{P}_X \subseteq \mathcal{P}_Y \iff X \leq_T Y, \quad \mathcal{L}_X \subseteq \mathcal{L}_Y \iff X \leq_T Y.$$

Thus equivalent Turing degrees give equal fields, and incomparable degrees give incomparable fields within each family.

Proof. All finite arithmetic, cover, Hensel, and finite-jet algorithms relativize. The ordinary-real algebraic proof also relativizes: finite isolating data and an X -computable strict-inequality search compute any fixed algebraic real root. The field and real-closure proofs therefore apply with $\mathbb{R}_X$. Countability follows from countably many oracle program indices for fixed X , and coefficient-zero extraction proves the real-trace assertion.

If $X \leq_T Y$, simulate each X query using Y , including cover queries. Conversely the constant $r_X = \sum_{n \geq 0} 2\chi_X(n)4^{-(n+1)}$ belongs to both fields for X . Inclusion in the corresponding field for Y makes r_X a Y -computable real. The separated-digit argument from Proposition 3.5 then recovers X from Y . This proves both equivalences. $\square$

Consequently each family contains strict chains indexed by $\emptyset, 0', 0'', \dots$. The exact degree embedding was explicit for Puiseux fields in $\mathbb{C}$; the left-finite statement here is its direct coefficient-and-cover analogue, with the proof given above.

Proposition 16.2 (Unrestricted oracle unions). *As X ranges over all subsets of $\mathbb{N}$, the union of the $\mathcal{P}_X$ is $\mathcal{P}(\mathbb{R})$, the full classical real Puiseux field. The union of the $\mathcal{L}_X$ is the full classical real Levi-Civita field.*

Proof. For a fixed classical Puiseux series, encode a countable joint table of fast rational names of its real coefficients into one oracle. For a fixed left-finite series, additionally encode a complete finite list of its support below each rational cutoff and choose one rational support lower bound. These are countable data, so one oracle suffices. The reverse inclusions are immediate from the respective support conditions. $\square$

The oracle union does not remove its support restriction. Even with unlimited oracle information, the Puiseux union misses the unbounded-denominator example in $\mathcal{L}_c$. Likewise no rank-one oracle change inserts ω^ω . Conversely enlarging supports cannot make a noncomputable real constant computable under unchanged ordinary Cauchy observations. Similar degree embeddings hold at every finite rank by the same real-trace argument. Matching the real trace of the structural field by allowing hyperarithmetic oracles does not prove equality of surreal fields or a name conversion.

16.1 General Hahn workspaces require effective decomposition data

There is another route to larger supports. Let G be a computable ordered abelian group with a specified ordered embedding into the surreal exponent group. A coefficient oracle on G can be supplemented by explicit finite *decomposition covers*, rather than by left-finite initial sections. The next observation states exactly the extra data required for one important operation.

Proposition 16.3 (An effective Neumann interface). *Let $h = \sum_{g \in G} a_g t^g$ have well-ordered positive support and uniformly computable real coefficients. Suppose a procedure, given g , returns a finite set $W(g)$ of ordered finite tuples $(g_1, \dots, g_n)$ with sum g , containing every such tuple whose entries all lie in $\text{supp}(h)$. The empty tuple is included at $g = 0$. Then for every uniformly computable real coefficient sequence (c_n) , the coefficients of $\sum_{n \geq 0} c_n h^n$ are uniformly computable.*

Proof. Neumann's support lemma gives the mathematical Hahn summability for well-ordered positive support [10]. Compute

$$[t^g] \left(\sum_{n \geq 0} c_n h^n \right) = \sum_{(g_1, \dots, g_n) \in W(g)} c_n a_{g_1} \cdots a_{g_n}.$$

Every actual nonzero contribution occurs in the list. Extra tuples have a zero coefficient factor and contribute zero. The list is a set, so there is no duplicate counting. The displayed expression is a finite real Cauchy computation. The empty product is one. $\square$

The classical lemma supplies existence and finiteness of actual contributions. The algorithm W supplies their effective completeness. One must not conflate those assertions. In a full named-workspace design, one must also supply whatever support certificates are required for output names and for subsequent operations. This proposition proves coefficient computation, not an automatic effective representation theorem for every Hahn field.

Theorem 5.1 demonstrates that even an apparently tame support can fail to supply the needed effective finite search information. The iterated construction supplies it by nested finite covers; a general Hahn implementation must expose its own equivalent mechanism.

17 Integration with the Surreal repository

17.1 Inspected revision and scope

The repository was inspected at revision

4896a2808ce30e01b1c86ae3ba2295a64762d246.

The reading included the root README and the files

```
Surreal/Foundations/SignSequenceField.lean
Surreal/HahnSeries/Evaluation.lean.
```

Repository structure information was also inspected [12]. This is a scoped source review, not a complete audit or a new build of the repository.

At this revision the sign-sequence carrier already has an ordered-field structure transported through its proved equivalence with numeric games. The field source explicitly separates the remaining genetic product-cut, real-closedness, and Hahn normal-form obligations. Thus it would be incorrect to describe multiplication and inversion on the mathematical carrier as wholly unimplemented. It would be equally incorrect to infer from those field instances that the numerical algorithms of this article have been implemented.

The Hahn evaluation source already requires positive order explicitly and proves the substituted-term interpretation, a geometric identity, and an exact finite remainder. Its evaluator is a mathematical, noncomputable Lean definition using existing Hahn infrastructure. This is the right semantic layer for our algorithms, but it does not supply computable finite-section covers or ordinary Cauchy coefficient procedures. The missing bridge is *effective data with correctness proofs*, not a second proof that a formal geometric sum exists.

17.2 A proof-carrying architecture

A practical design should keep three layers separate.

Semantic values. Use the existing Hahn field as the primary semantic target for the rank-one development. Prove the properties of $\mathcal{L}_c$ first there. Transport them into the sign-sequence surreal carrier only after the normal-form embedding is available. A theorem about a named Hahn series must not be advertised as a compiled theorem about the full surreal carrier before that bridge is proved.

Executable names. Represent the rational lower bound, the coefficient program, and the finite-cover program as explicit data. A validity proposition relates those programs to a semantic series. The representation should not force a canonical nonzero-support list, since that would hide a real zero test. For an initial executable prototype, exact rational or algebraic coefficients are easier; ordinary Cauchy coefficients must later retain the nonuniformity distinctions proved here.

Operation certificates. Expose separate certificates for positive support, unit leading term, leading exponent, a simple residue root, and a valuation-Cauchy modulus. These certificates express different facts and cannot safely be collapsed into a single “nonzero” flag. In particular, a positive-support proof can justify deleting nonpositive zero candidates, and a unit proof can justify inversion with leading exponent zero. Both are useful even when a general normalization procedure is impossible.

The following is a data-contract sketch, not compiled Lean code:

```

NumericalName:
  lower : Rational
  coefficient(q, precision) : Rational
  cover(bound) : finite set of Rational
  validity : error bounds, lower support bound,
             and completeness of every finite cover

LeadingCertificate(name):
  exponent : Rational
  coefficient_at_exponent_is_nonzero
  all_earlier_coefficients_are_zero

PositiveSupportCertificate(name):
  all_nonpositive_coefficients_are_zero

HenselInput:
  integral_polynomial_names
  residue_root_name
  residue_root_equation
  residue_derivative_is_nonzero

```

An output certificate may be proved without being found by a general search. For example, differentiating a formal series proves that its t^{-1} coefficient is zero; multiplying a unit by another unit gives a unit certificate; and the Hensel invariant supplies the derivative's leading coefficient at every Newton stage. Proof generation along these specific constructions is much more realistic than universal normalization.

17.3 A dependency order for formalization

The first layer should formalize finite covers, truncation, and the two finite formulas (7.1)–(7.2). These are finitary facts independent of real closedness. Next come the positive-gap lemma, effective geometric series, and certified inverse, with a clear separation between program construction and semantic correctness. Effective stabilization and simple Hensel lifting then provide the main analytic-algebraic engine. Real closedness can be proved through an available valuation criterion or through an explicitly formalized version of Lemma 8.3; it should not be added as an unproved instance.

The negative results are valuable formalization targets as well. The reduction using $a_e + t$ is much smaller than a full computable-analysis library and guards against designing an impossible API. The support-product counterexample tests a different assumption and should remain separate. Structural validity and the hyperarithmetic classification belong to a higher-recursion-theory track, rather than being prerequisites for the rank-one numerical algorithms.

No new Lean files are claimed to compile in this package. The supplied Python suites check finite exact identities and constructions only. A source-level proof in this article, an executable test, and a kernel-checked Lean theorem are three different kinds of evidence.

18 Comparison, boundaries, and further research

18.1 A capability comparison

Feature	Structural field $\mathcal{S}_{\text{str}}$	Numerical core $\mathcal{L}_c$
Input meaning	Hereditary effective cut construction	Cauchy coefficients and finite support covers
Real slice	Hyperarithmetical reals	Ordinary computable reals
Scales	All computable ordinal scales, and more values	Rational powers of ω ; enlarged by the atlas
Equality	Not a general decision procedure	Not decidable
Ring operations	Uniform cut compilers	Uniform finite-cover algorithms
Field closure	Established extensional theorem	Proved here; inverse names need normalization data uniformly
Root theory	Established real closedness	Effective simple Hensel lifts and extensional real closedness
Convergence	No nontrivial small net in full-surreal topology	Effective valuation limits with a modulus

The entry for structural scales does not claim that every surreal of computable birthday has an effective structural presentation. Likewise, the atlas is not asserted to exhaust structural computability. A complete effective translation between arbitrary structural names, ordinal sign presentations, and the numerical names above is not proved here. None of the main numerical theorems relies on such a translation.

18.2 What is established, and what is not

The main theorem has now been proved. Its countability and real slice are Proposition 6.2; uniform ring arithmetic is Theorem 7.1; certified and extensional inversion are Theorem 7.4; the precise obstruction is Theorem 12.1; Hensel lifting and real closedness are Theorems 8.1 and 8.4; Puiseux closure and effective completion are Theorems 9.3 and 11.3; and the functional and surcomplex claims are Theorem 13.1, Theorem 13.3, and Proposition 14.1. The iterated atlas is an additional extension, proved in Theorem 15.1.

Several tempting stronger assertions were deliberately excluded. We did not prove a general uniform algebraic root-isolation algorithm on raw Cauchy–Hahn names. We did not define a universal exact symbolic zero test. We did not infer effective finite decomposition from abstract Hahn summability. We did not confuse a computable copy of a birthday with computable rank access to its signs. We did not claim that valuations, coefficient approximation, and full-surreal neighborhoods describe the same topology. Finally, the current normal-form bridge and the new computability theory have not been machine-checked as part of this work.

18.3 Research directions with precise targets

The following are directions left by this article, not claims that the corresponding questions have never been studied elsewhere.

Uniform translation complexity. Determine the exact information needed to convert a numerical normal-form name into an effective sign presentation, and conversely on a stated subclass. The

computable-copy example shows that rank access must be part of the specification. The structural classification supplies a natural upper-level language, but does not by itself yield a practical name translator.

Sharper operator degrees. The normalization and inverse tasks have halting degree in the computable index setting. Their precise represented-space or Weihrauch classification under different cover encodings may distinguish one zero test from finite parallel tests or more elaborate search. Any such refinement should preserve the elementary lower-bound family $a + t$.

Support certificates beyond left-finiteness. Design reusable certified decomposition structures for more general computable ordered groups, especially supports with genuine transfinite order type rather than finite-rank nested left-finite structure. A useful target is closure under multiplication and geometric inversion with explicit finite dependency output for each requested coefficient. Theorem 5.1 is a minimal adversarial test.

Effective algebraic branch data. Extend the certified algorithms already proved here to broader informative root data: a specified Newton slope, a simple initial root, a ramification bound, or a certified factorization pattern. The simple Hensel theorem and the finite-jet branch certificate are already successful contracts. Repeated roots should not be handled by assuming a zero decision for arbitrary coefficients.

Larger transcendental workspaces. Extend the numerical atlas by explicitly named logarithmic and exponential scales, while proving that the real slice remains $\mathbb{R}_c$ and specifying finite dependency rules for all included operations. A finite symbolic expression and an effective coefficient expansion are different output formats; a system can support both without asserting that they are uniformly interconvertible.

18.4 Conclusion

A workable concept of computable surreal number is necessarily representation-sensitive. Structural presentations capture hereditary construction and admit hyperarithmetic real information. Numerical Cauchy–Hahn names capture effective observation of coefficients and their support dependencies, recover exactly the ordinary computable reals on the real axis, and support a substantial real-closed theory.

The constructive side is not small: the numerical fields contain infinitesimals, infinite numbers, arbitrary computable formal coefficient families on controlled supports, algebraic roots, effective valuation limits, and a hierarchy of higher scales. The obstructions are equally precise. Finite mathematical dependency does not imply an effective list; field closure does not imply a uniform inverse compiler; and a finite program for a transfinite structure does not necessarily compute its real value. Respecting those distinctions produces algorithms whose mathematical contracts can actually be met.

A Finite algorithms and reproducibility

A.1 Rerun checks from all three reports

All three original verification programs are retained unchanged, alongside a new cross-model suite. They use Python standard-library exact arithmetic without network access. Rerun results:

Suite	Reported unit	Rerun result
A	4,967 exact comparisons	PASS
B	18 named unit tests	PASS
C	266 exact checks	PASS
Merged	4,321 cross-model exact checks	PASS

Unlike test units are not added together. The merged suite compares dense and sparse convolution, redundant finite covers, branch transforms, residue identities, and denominator examples. Synthetic halting fixtures do not compute the halting set.

These finite checks do not prove the infinite theorems or validate arbitrary coefficient programs. No new Lean verification or production surreal computer-algebra implementation is claimed.

From the package root:

```
python3 code/run_all.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

A.2 A precise finite-dependency recipe

For reference, the positive-support evaluation algorithm can be implemented by the following mathematical pseudocode. All polynomial coefficients below are real *names*, and `finite_arithmetic` invokes the correct real precision subroutine rather than replacing those coefficients by exact floating-point numbers.

```
coefficient_of_evaluation(f, h, q, precision, positive_support_proof):
    candidates = h.cover(1)
    delta = min({1} union {r in candidates : r > 0})
    choose N >= 0 with (N + 1) * delta > q
    form the named finite sum S = sum(n = 0..N, f[n] * h^n)
    return S.coefficient(q, precision)

cover_of_evaluation(f, h, bound, positive_support_proof):
    extract delta as above
    form the finite set of n >= 0 with n * delta < bound
    for these n, compute covers of f[n] * h^n below bound
    return their union, filtered to [0, bound)
```

For inversion, first normalize using the *supplied* leading exponent and its nonzero coefficient, then call this recipe with $f_n = 1$. For Hensel lifting, every derivative is a certified unit, so its leading exponent is zero without a normalization search. These are explicit finite computations per output request, even though a whole infinite series is never printed.

B Reconciliation ledger and source crosswalk

This appendix records the mathematical disposition of the three manuscripts. Locations below use their original TeX line numbers, not the repaginated PDF. A has 2,235 lines, B has 1,916, and C has 2,221. The three source manuscripts are *not* redistributed with this report; the line numbers and the recorded hashes identify them, and the disposition of every result is set out below, but the texts themselves are not included here. The JSON manifest records full hashes; upload suffixes alone are not treated as version history. A supplied the continuous left-finite development; shared introductions, repeated proofs, bibliographies, and implementation advice were consolidated rather than concatenated.

B.1 Actual corrections and necessary qualifications

Nonzero polynomial hypothesis. A, the corollary immediately after its real-closedness theorem (source lines 1083–1093), says every real surreal root of a polynomial over $\mathcal{L}_c$ is in $\mathcal{L}_c$. Literally this fails for the zero polynomial. The merged statement explicitly requires a *nonzero* polynomial, as C already does. This is a corrected hypothesis, not merely a notation change.

Two fields, not two answers about one field. A's $\mathcal{L}_c$ includes effective finite covers and variable rational denominators; B–C's $\mathcal{P}_c$ has one bounded denominator per value. The proper inclusion and completion theorem reconcile their scope. A's effective completeness is never used to negate B–C's Puiseux incompleteness, and is not upgraded to unqualified sequential completeness of $\mathcal{L}_c$.

A source question is answered by another source. C's first concluding question asks for effective supports beyond bounded Puiseux denominators with algebraic closure and finite dependency data. The finite-cover left-finite construction answers the real-closed version and its complexification answers the algebraically closed version. The remaining support research target is moved beyond that established case.

Closure is not a uniform algorithm. Every proof that chooses a leading exponent, multiplicity, finite jet, or coefficient inverse is labeled as either finite nonuniform existence data or an additional algorithmic input. No claim of raw-name field computability is inferred from real closedness. C's positive-square-root lower bound and B's any-root quadratic lower bound are retained separately.

Two different uses of one jump. B's unrestricted convolution algorithm returns an oracle-computable coefficient function; it need not yield an ordinary computable product. A–C's inverse and square-root normalization uses $0'$ to choose finite data, after which an ordinary program suffices. The convolution search is phrased with a c.e. contributor set, not an unjustified decidable zero test on real coefficients. All such statements concern valid ordinary computable indices; relative names require the corresponding relative jump.

Support and valuation conventions. All three use $t = \omega^{-1}$; increasing t exponents correspond to decreasing ω exponents. Candidate covers may include zeros and need not be nested. A finite

cardinality bound is not a finite candidate list. At higher rank, full natural valuation and outer valuation are distinguished. The formal residue $[t^{-1}]x$ is not standard part $[t^0]x$.

Limits, certificates, and historical status. Exact coefficient-name jets and rational coefficient approximations are separate observations. Exact valuation stabilization and real coefficientwise Cauchy moduli are separate limit hypotheses. Proof-bearing certificates do not make arbitrary semantic validity decidable. The structural classification is credited to established work, not re-announced as a new theorem; the repository’s remaining real-closedness task means a formalization obligation, not an open question about the classical surreal field.

B.2 Topic-by-topic disposition

Topic	Original locations	Merged disposition
Real names, size and topology	A 223–318; B opening conventions; C 242–401	One common fast-Cauchy baseline, intrinsic versus ambient topology, and a separate note on infinitary machines.
Structural field and halting real	A 319–501; B structural section; C 402–666	Common cut compiler and separated base-four encoding retained; classification imported with coding caveat.
Validity and birthdays	A 478–500; C 480–547	Full Π_1^1 upper bound retained from C; all-left reduction supplies hardness. Recursive-ordinal boundary made explicit.
Signed computable orders	A 502–556; B signed-order construction; C 667–727	C’s theorem for every c.e. set retains exact Turing degree; the finite-block variants are consolidated.
Hahn convolution failures	A 557–643; B convolution and one-jump section; C 728–821	Both left-finite and singleton-fiber examples retained; B’s dyadic template recorded; sharp $0'$ upper bound included.
Finite-cover numerical model	A 644–902; C convolution certificate	A’s definition and full arithmetic proofs retained. Complete finite candidate lists are the operational requirement.
Hensel and valuation proof	A 903–1094	Retained with the explicit defectless residue-characteristic-zero input; zero polynomial excluded in the root corollary.
Puiseux field and inclusion	A 1095–1192; B Puiseux field; C 822–1028	One definition, common-denominator ring formulas, certified reciprocal, exact trace, and proper inclusion.
Finite advice and algebraic branches	B simple Hensel and finite-advice normalization; C 1029–1236	One full finite-jet proof, independent of the valuation proof, with Laurent reduction and an explicit certificate.
Uniformity and zero tests	A 1295–1479; B root/sign obstructions; C 1237–1428	Strongest compatible lower bounds; full promised-index classification; exact-coefficient alternative retained.
Limits and completion	A 1193–1294; C 1556–1643	Effective completion in $\mathcal{L}_c$, failure in $\mathcal{P}_c$, and a separate coefficientwise real limit theorem.
Formal functions and equations	A 1480–1610; B differentiation/integration; C 1429–1555	Formal evaluation, implicit examples, residue quotient, and a differential non-solvability example; not global analysis.

Topic	Original locations	Merged disposition
Surcomplex	A 1611–1662; B complex section; C 1840–1891	Both complexifications; uniform squared modulus versus nonuniform modulus; branch-dependent operations qualified.
Finite rank	A 1663–1776; B iterated Puiseux construction; C 1644–1750	Parallel $\mathcal{P}_d$ and $\mathcal{L}_d$ towers with proper inclusions, correct residues, and a single reused $0'$.
Oracle hierarchies	C 1751–1839	Puiseux degree embedding retained and proved for left-finite names too; unrestricted oracle unions identified separately.
General supports and development plan	A 1777–2043; B support certificates and repository plan; C 1892–2044	General Neumann certificates retained; redundant-name API, branch contract, and remaining support questions unified.
Checks and proof provenance	A 2064–2235; B verification/bibliography; C 2045–2221	All original test suites rerun; counts kept separate; additional cross-model tests; original source hashes preserved.

B.3 Bibliographic and repository checks

The structural definition and classification were checked directly in pages 4–6 of Hamkins’s 2024 lecture notes and his 2025 announcement. Lurie’s publisher record confirms June 1998, volume 63(2), pages 337–371; its 2014 online posting is not a new publication date. The full Lurie and Harkleroad papers were not independently read in this merge. The Newton–Puisseux input was checked in the introductions of Rond and Kedlaya; the defect formula was checked in Rzepka–Szewczyk, Section 2.1, equation (1). The classical left-finite definition was checked in Restrepo Borrero–Shamseddine, Section 1.1, version 2. None of those sources is cited as having proved the particular representation-specific synthesis above.

The repository tree and selected files were read at commit

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the same snapshot recorded by the reports. The root README, the sign-field source, and the Hahn evaluation source agree on the distinction between mathematical infrastructure and effective observation algorithms. No new commit, repository write, or local Lean build is part of this deliverable.

C A proof-dependency ledger

Result	Dependencies and status
Structural classification	External theorem; historical sources and modern authors' statements cited. Not reproved.
Structural halting real	Effective cut and ring construction; independent explicit reduction, with prior phenomenon acknowledged.
Computable-copy sign trap	Finite halting simulations, canonical real sign expansions, and a direct decoding argument.
Noncomputable Hahn product	Explicit rational supports and base-four uniqueness. Full proof provided.
Numerical ring and field	Definition of covers; finite Cauchy arithmetic; positive gap; geometric identity. Full proofs provided.
Effective Hensel lifting	Certified unit inversion; finite Taylor identity; explicit valuation doubling and coefficient stabilization. Full proof provided.
Numerical real closedness	Hensel lifting, the proved residue and value-group descriptions, and the explicitly cited Ostrowski degree formula.
Puiseux real closedness	Two independent proofs: the common-grid Hensel/valuation route and Newton–Puiseux plus the proved finite-jet uniformity step.
Effective completion	Exact stabilization modulus; finite covers from a selected sequence term. Full proof provided.
Uniform inverse obstruction	The uniformly Cauchy-computable real family a_e and the t^{-1} coefficient of $(a_e + t)^{-1}$. Full proof provided.
Higher-rank atlas	Nested finite covers, one fixed coefficient-inverse choice per Hensel problem, and induction on rank. Full proof provided.
Oracle hierarchies	Relativized algorithms and the explicit separated-digit real encoding; both support classes treated.
Formalization status	No new Lean verification claimed; finite Python tests are supplementary only.

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